

Elementary acoustic sources

□ Elementary acoustic sources can be solved in two different ways:

➤ **Inhomogeneous wave equation**

- Sources are introduced *locally* either in the form of a mass flow q or in the form of a volume force $\mathbf{F}$
- Involves **Green functions** (will be discussed later...)

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = \frac{\partial q}{\partial t} - \frac{\partial F_i}{\partial x_i} = Q(\mathbf{x}, t)$$

or

$$\frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = -i\omega q - \frac{\partial F_i}{\partial x_i} = Q(\mathbf{x}, \omega)$$

➤ **Homogeneous wave equation**

- Solution is achieved by setting boundary conditions to determine the unknown constants introduced by the sources
- **Will be discussed next...**

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \frac{\partial^2 p'}{\partial x_i^2} = 0$$

or

$$\frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = 0$$

Case 1 - Pulsating sphere

- Consider the sound radiation from a small sphere of radius a , whose surface oscillates radially with a normal surface velocity $[v_r]_{r=a}$, pulsating with ω

$$\text{BC: } \begin{cases} v_r = v_0 e^{-i\omega t}, & \hat{v}_r = v_0, & r = a \\ \frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = 0, & & r \geq a \end{cases}$$

- The solution to the wave equation matching the BC is:

$$\hat{p}(r) = \frac{\hat{A} e^{ikr}}{r}$$

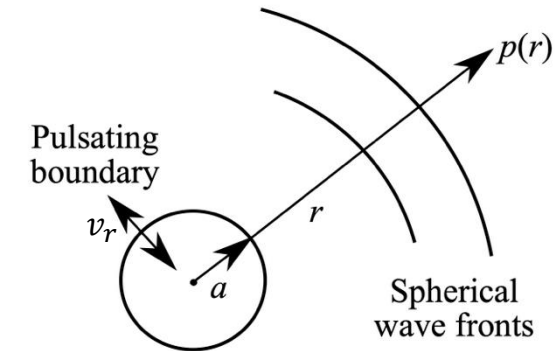
$$\hat{v}(r) = \frac{\hat{p}(r)}{z_0} \left(1 + \frac{i}{kr} \right)$$

- At $r = a$ we have: $[\hat{v}]_{r=a} = v_0 = \left[\frac{\hat{p}(r)}{z_0} \left(1 + \frac{i}{kr} \right) \right]_{r=a} \Rightarrow v_0 = \frac{\hat{A} e^{ika}}{a z_0} \left(1 + \frac{i}{ka} \right) \Rightarrow$

$$\hat{A} = - \frac{i \omega \rho_0 a^2 v_0 e^{-ika}}{(1 - ika)}$$

- The **acoustic field** is thus:

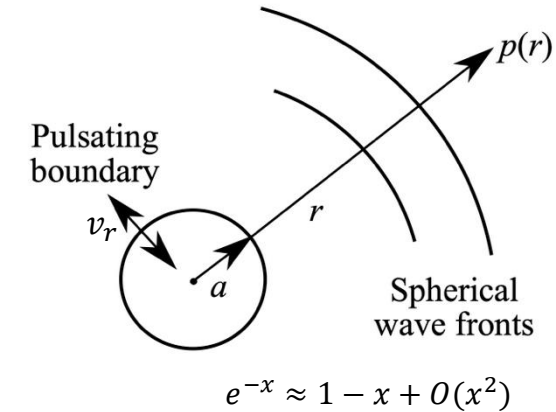
$$\hat{p}(r) = - \frac{i \omega \rho_0 a^2 v_0}{(1 - ika)} \frac{e^{ik(r-a)}}{r}$$



Case 1 - Pulsating sphere

- If the sphere is acoustically compact ($ka \ll 1$), we approximate: $e^{-ika} \approx 1 - ika$
- Therefore, the solutions approximates to:

$$\hat{p}(r) = -\frac{i\omega\rho_0 a^2 v_0 e^{ikr}}{r} \propto \frac{1}{r}$$



- **Note:** no initial conditions were required because a *harmonic time dependence* was assumed
- Given a sphere's surface area of $S = 4\pi a^2$, we can define the **rate of change of the sphere's volume** (also known as its **volumetric flow rate**):

$$Q = \int_S [v_r]_{r=a} dS = \int_0^{2\pi} \int_0^\pi v_0 e^{-i\omega t} a^2 \sin \theta d\theta d\varphi \Rightarrow$$

$$\begin{aligned} Q(t) &= v_0 S e^{-i\omega t} \\ \hat{Q} &= v_0 S \end{aligned}$$

$$M(t) = \rho_0 \hat{Q} e^{-i\omega t}$$

mass flow rate

- **Monopole source**
Volume displacement source
(*Omni-directional*)

$$\hat{p}(r) = -i\omega\rho_0 \hat{Q} \frac{e^{ikr}}{4\pi r}$$

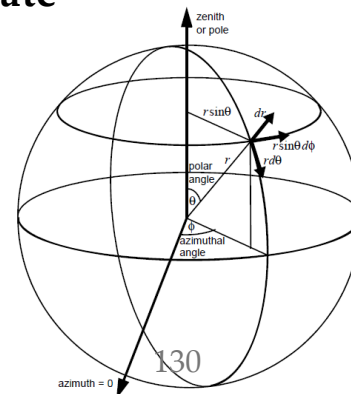
Harmonic monopole
source strength

$$\hat{v}(r) = -ik\hat{Q} \frac{e^{ikr}}{4\pi r} \left(1 + \frac{i}{kr}\right)$$

Free-field Green's function

$$G = \frac{e^{ikr}}{4\pi r}$$

Aircraft Aeroacoustics

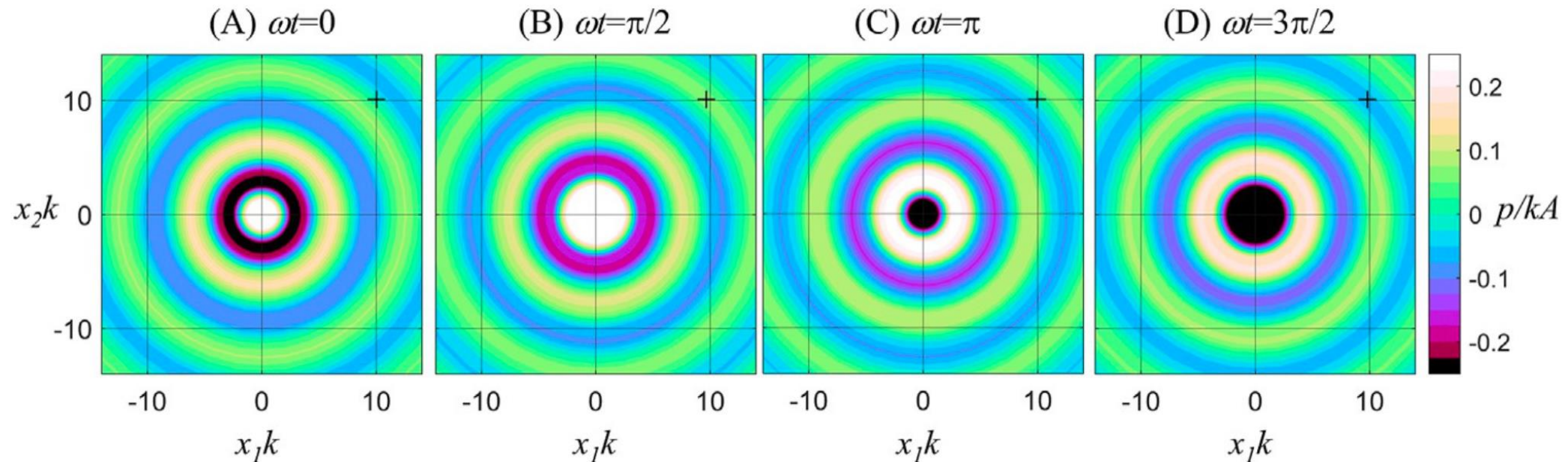
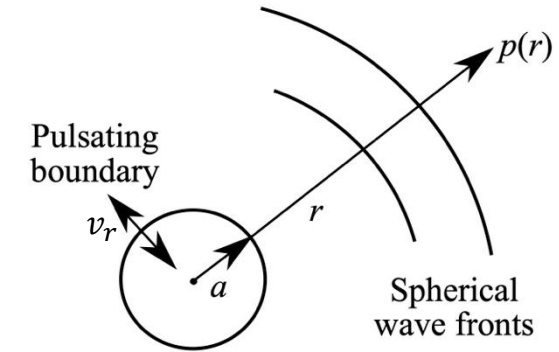


Case 1 - Pulsating sphere

□ Monopole source

$$\frac{\hat{p}(r)}{k\hat{A}} = \frac{e^{ikr}}{kr}$$

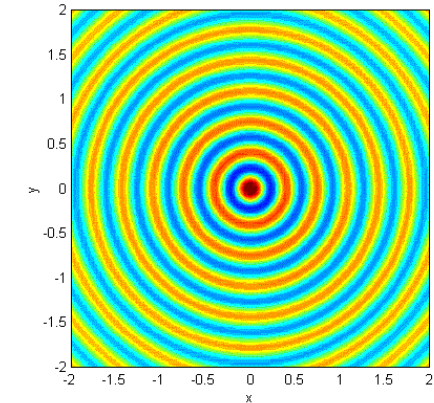
$$p'(r, t) = \Re[\hat{p}(r)e^{-i\omega t}]$$



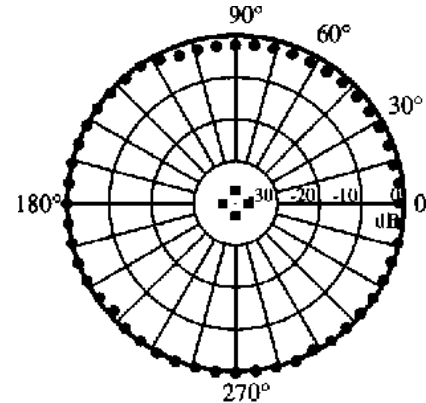
Monopole noise sources

□ Examples

$$\hat{p}(r) = -i\omega\rho_0\hat{Q}\frac{e^{ikr}}{4\pi r}$$



- **Boxed loudspeaker** acts as an omnidirectional monopole source at low frequencies

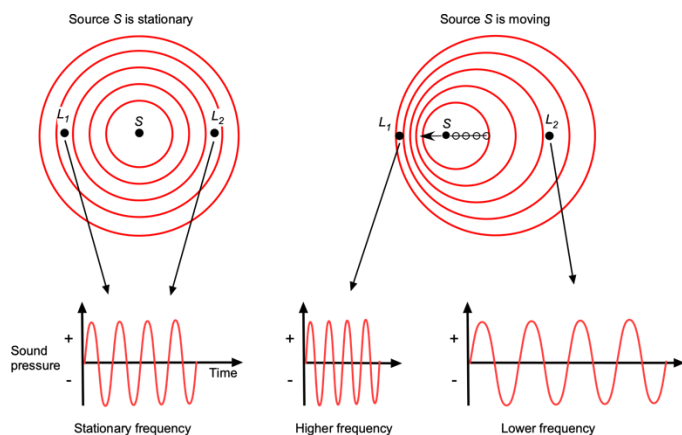
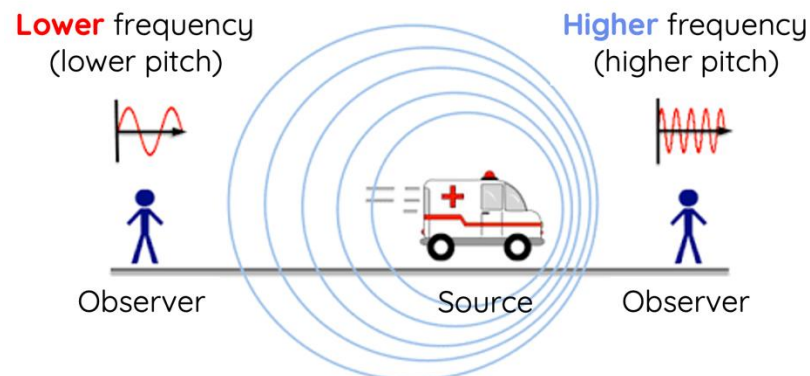


Measured directivity patterns at 250 Hz for sound radiation from a monopole 4" [boxed loudspeaker](#)

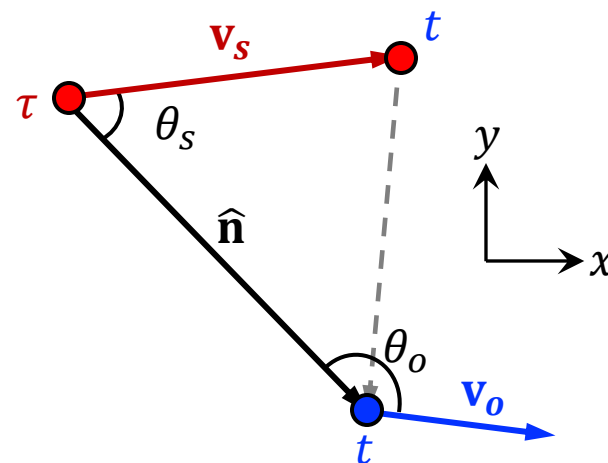
Monopole noise sources

Examples

Siren



$$\hat{p}(r) = -i\omega\rho_0\hat{Q}\frac{e^{ikr}}{4\pi r}$$



$$f' = \frac{c_0 - \mathbf{v}_o \cdot \hat{\mathbf{n}}}{c_0 - \mathbf{v}_s \cdot \hat{\mathbf{n}}} f = \frac{c_0 + |\mathbf{v}_o| \cos \theta_o}{c_0 - |\mathbf{v}_s| \cos \theta_s} f$$

$|\mathbf{v}_o| \cos \theta_o \rightarrow$ negative when *observer* is moving away the *source*

$|\mathbf{v}_s| \cos \theta_s \rightarrow$ negative when *source* is moving away from the *observer*

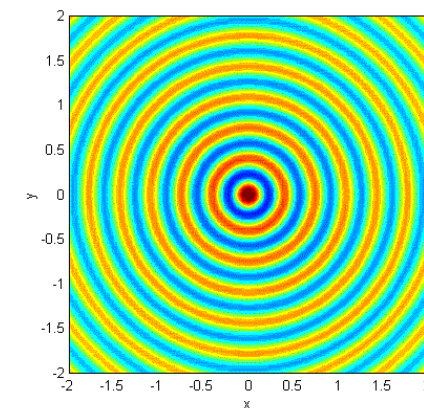
$\hat{\mathbf{n}}$ – unit vector from *source* to *observer* at the emission time

$\mathbf{v}_o \rightarrow$ *observer* velocity vector

$\mathbf{v}_s \rightarrow$ *source* velocity vector

τ – sound source emission time

t – observer time when hearing the sound emitted at time τ



Case 2 - Translating sphere

- Consider a sphere of radius a that translates back and forth in the x_1 direction with velocity v_1 , pulsating with ω
- Viscous effects are ignored $\rightarrow$ **only the radial velocity of the surface of the sphere is considered** (tangential velocity to the surface is neglected)

$$\text{BC: } \begin{cases} v_r = v_0 \cos \theta e^{-i\omega t}, & \hat{v}_r = v_0 \cos \theta, & r = a \\ \partial^2 \hat{p} / \partial x_i^2 + k^2 \hat{p} = 0, & & r \geq a \end{cases}$$

- The solution to the wave equation matching the BC is:

$$\hat{p}(r) = \frac{\partial}{\partial x_1} \left(\frac{\hat{A} e^{ikr}}{r} \right)$$

Note: any derivative of a solution to the wave equation is also a solution to the wave equation!

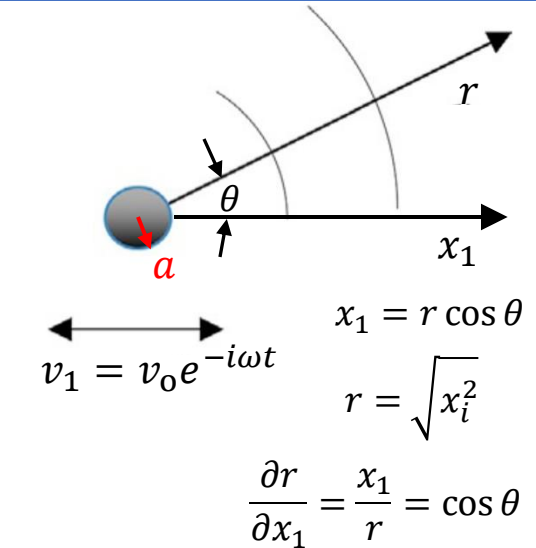
- Evaluating the derivative above yields: $\hat{p}(r) = \frac{\partial r}{\partial x_1} \frac{\partial}{\partial r} \left(\frac{\hat{A} e^{ikr}}{r} \right) \Rightarrow$

$$\hat{p}(r) = ik \cos \theta \left(\frac{\hat{A} e^{ikr}}{r} \right) \left(1 - \frac{1}{ikr} \right)$$

- Note** – near-field sound amplitude is *proportional* to $\hat{p} \propto \frac{1}{r} \left(1 - \frac{1}{ikr} \right)$

➤ In acoustic far-field ($kr \gg 1$), we obtain the regular $\hat{p} \propto 1/r$:

$$kr \gg 1 \Rightarrow \hat{p}(r) = ik \cos \theta \left(\frac{\hat{A} e^{ikr}}{r} \right)$$



Directivity term
can be used to match BC
for the translating sphere

Case 2 - Translating sphere

- From the acoustic momentum conservation, we obtain: $\hat{\mathbf{v}} \cdot \mathbf{n}_r = \frac{1}{i\omega\rho_0} \frac{\partial \hat{p}}{\partial r}$
 $\triangleright \mathbf{n}_r$ – unit vector in the radial direction on the sphere's surface

- The BC at sphere surface states: $[\hat{\mathbf{v}} \cdot \mathbf{n}_r]_{r=a} = [\hat{v}_r]_{r=a} = v_0 \cos \theta$

- Thus, substituting the BC into the momentum conservation yields:

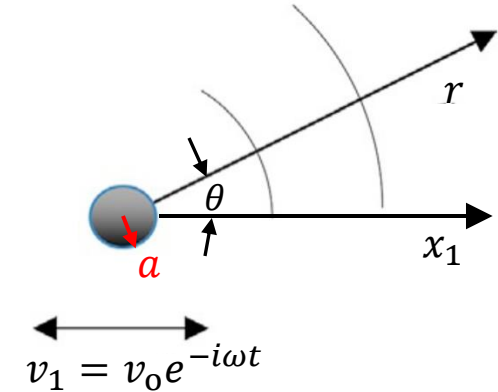
$$[\hat{\mathbf{v}} \cdot \mathbf{n}_r]_{r=a} = \left[\frac{1}{i\omega\rho_0} \frac{\partial \hat{p}}{\partial r} \right]_{r=a} \Rightarrow v_0 \cos \theta = \frac{1}{i\omega\rho_0} \left[\frac{\partial \hat{p}}{\partial r} \right]_{r=a}$$

$$\Rightarrow v_0 \cos \theta = \frac{\cos \theta}{c_0 \rho_0} \left(\frac{\hat{A} e^{ika}}{a} \right) \left(ik - \frac{2}{a} + \frac{2}{ika^2} \right) \xRightarrow{\text{solving for } \hat{A}} \boxed{\hat{A} = ika^3 c_0 \rho_0 v_0 \left(\frac{e^{-ika}}{2 - (ka)^2 - 2ika} \right)}$$

- If the **sphere is acoustically compact** ($ka \ll 1$), we find that:

$$\hat{A} = ika^3 c_0 \rho_0 v_0 \left(\frac{e^{-ika}}{2 - (ka)^2 - 2ika} \right) \Rightarrow \hat{A} \approx \frac{1}{2} ika^3 c_0 \rho_0 v_0 e^{-ika} \xRightarrow{e^{-ika} \approx 1 - ika} \hat{A} \approx \frac{1}{2} ika^3 ika c_0 \rho_0 v_0 \left(\frac{1}{ika} - 1 \right)$$

$$\Rightarrow \boxed{\hat{A} \approx \frac{1}{2} ika^3 c_0 \rho_0 v_0}$$



$$\hat{p}(r) = ik \cos \theta \left(\frac{\hat{A} e^{ikr}}{r} \right) \left(1 - \frac{1}{ikr} \right)$$

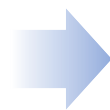
$$\frac{\partial \hat{p}}{\partial r} = ik \cos \theta \left(\frac{\hat{A} e^{ikr}}{r} \right) \left(ik - \frac{2}{r} + \frac{2}{ikr^2} \right)$$

Case 2 - Translating sphere

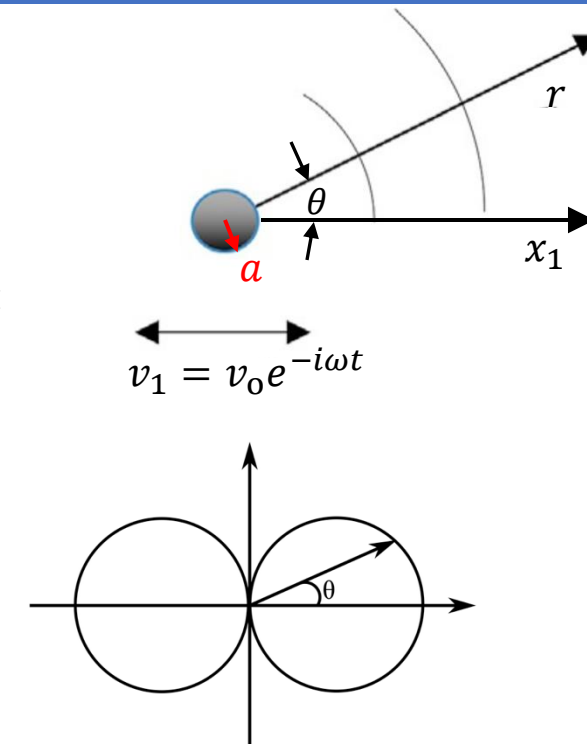
□ Substituting $\hat{A}$ into the $\hat{p}(r)$ expression results in the following acoustic field:

$$\hat{A} \approx \frac{1}{2} i k a^3 c_0 \rho_0 v_0 \Rightarrow \hat{p}(r) = i k \cos \theta \left(\frac{\hat{A} e^{i k r}}{r} \right) \left(1 - \frac{1}{i k r} \right)$$

□ Dipole source



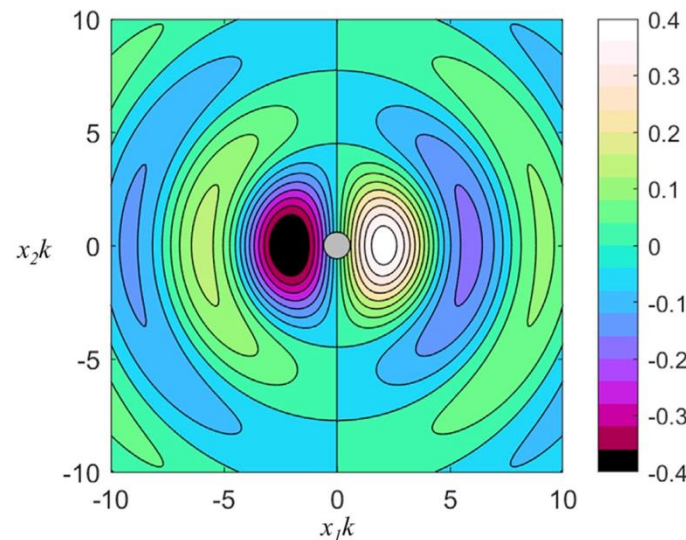
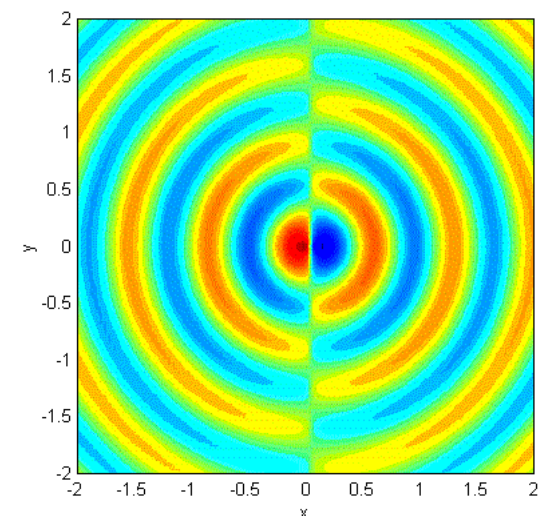
$$\hat{p}(r) = \frac{i k a \cos \theta}{2} \left(\frac{i \omega \rho_0 v_0 a^2 e^{i k r}}{r} \right) \left(1 - \frac{1}{i k r} \right)$$



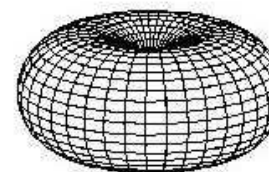
Directivity term – $\cos \theta$

Maximum amplification of the acoustic field along the x_1 -axis

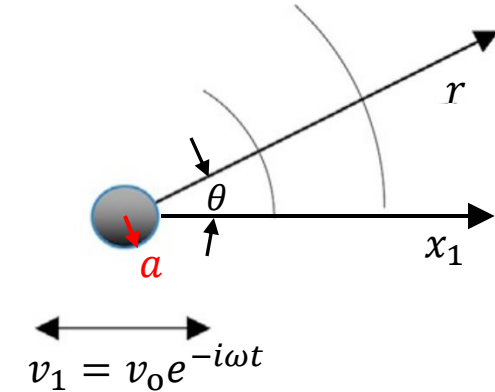
Zero amplification along the dipole principle axis



$$\text{Re} \left\{ \frac{2 \hat{p}}{\omega \rho_0 v_0 a^3 k^2} \right\}$$



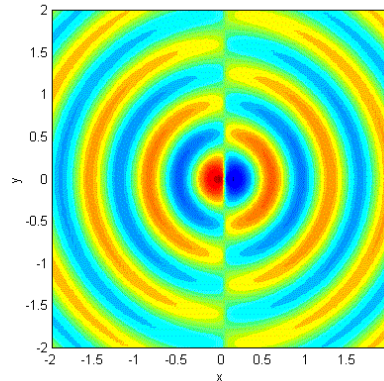
Case 2 - Translating sphere



- Pressure amplitude of translating sphere (**dipole**) is less than a pulsating sphere (**monopole**) by a factor of $ka/2$
 - **Pulsating sphere displaces mass** during each cycle, and so the medium has nowhere to go apart from propagating away as an acoustic wave
 - **Translating sphere causes no net displacement of mass**, and the fluid can adjust in the near field to accommodate the motion; yet some energy still escapes as sound and propagates to the acoustic far field

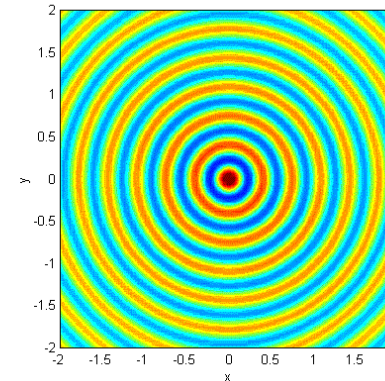
Dipole source

$$\hat{p}(r) = \frac{ika \cos \theta}{2} \left(\frac{i\omega \rho_0 v_0 a^2 e^{ikr}}{r} \right) \left(1 - \frac{1}{ikr} \right)$$



Monopole source

$$\hat{p}(r) = -\frac{i\omega \rho_0 a^2 u_0 e^{ikr}}{r}$$



Case 2 - Translating sphere

□ What about the **force** needed to move the translating sphere?

- For the *translating sphere*, the surface pressure depends on $\cos \theta \Rightarrow$ **spatial variation**
- The **total force** $\hat{\mathbf{F}}$ applied to the fluid will be the **sum of**:
 - Force required to **move the mass of fluid displaced by the sphere** ($m = \rho_o V = \rho_o \frac{4}{3} \pi a^3$)
 - Force required to **overcome the net surface pressure** $\hat{p}$

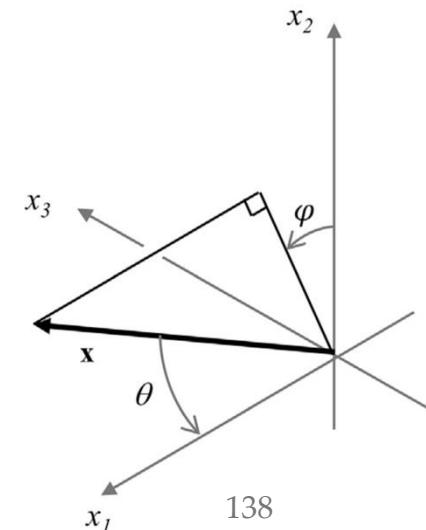
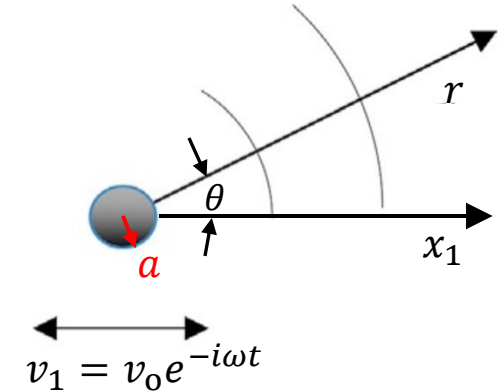
$$\hat{\mathbf{F}} = \int_V \rho_o \left(\frac{d\mathbf{v}}{dt} \right) dV + \int_S [\hat{p}]_{r=a} \mathbf{n}_r dS$$

Note: no net force was required for the *pulsating sphere* since the surface pressure is constant over the sphere's surface and integrates to zero

- Recognizing that: $\frac{d\mathbf{v}}{dt} = -(i\omega \hat{\mathbf{v}}) e^{-i\omega t}$ and $dV = \int_V r^2 \sin \theta dr d\theta d\varphi = \frac{4}{3} \pi a^3$

we can therefore write:

$$\Rightarrow \hat{\mathbf{F}} = -(i\omega \hat{\mathbf{v}}) \rho_o \left(\frac{4\pi a^3}{3} \right) + \int_S [\hat{p}]_{r=a} \mathbf{n}_r dS$$



Case 2 - Translating sphere

- What about the **force** needed to move the translating sphere?

$$\hat{\mathbf{F}} = -(i\omega\hat{\mathbf{v}})\rho_0\left(\frac{4\pi a^3}{3}\right) + \int_S [\hat{p}]_{r=a} \mathbf{n}_r dS$$

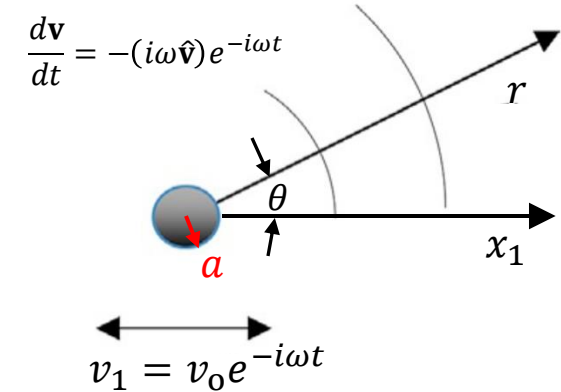
- The **only force to develop will be in the direction of the sphere's motion** (x_1), and so we can limit our analysis to that direction only
- In spherical coordinates we have $dS = a d\theta a \sin \theta d\varphi = a^2 \sin \theta d\theta d\varphi$, and so:

$$\hat{F}_1 = -i\omega\rho_0 v_0 \left(\frac{4\pi a^3}{3}\right) + \int_0^\pi \int_0^{2\pi} ([\hat{p}]_{r=a} \cos \theta) a^2 \sin \theta d\theta d\varphi$$

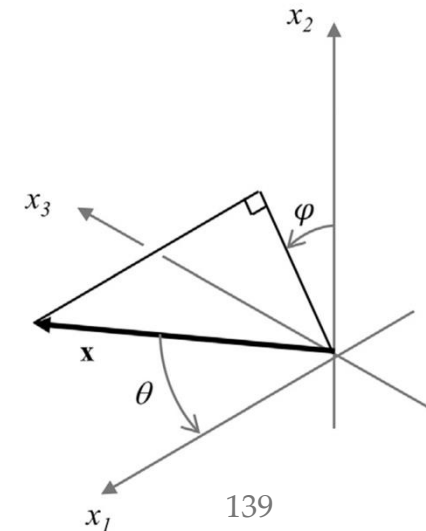
- Where at $r = a$ for acoustically compact sphere ($ka \ll 1$):

$$[\hat{p}]_{r=a} = \left[\frac{ika \cos \theta}{2} \left(\frac{i\omega\rho_0 v_0 a^2 e^{ikr}}{r} \right) \left(1 - \frac{1}{ikr} \right) \right]_{r=a} \Rightarrow \boxed{[\hat{p}]_{r=a} = -\frac{i\omega\rho_0 v_0 a \cos \theta}{2}}$$

$e^{ika} \approx 1$



$$\hat{\mathbf{v}} = v_0 \mathbf{x}_1$$



Case 2 - Translating sphere

□ What about the **force** needed to move the translating sphere?

➤ We therefore obtain:

$$\hat{F}_1 = -i\omega\rho_0 v_0 a^3 \left\{ \left(\frac{4\pi}{3} \right) + \pi \int_0^\pi (\cos^2 \theta) \sin \theta d\theta \right\} \Rightarrow \boxed{\hat{F}_1 = -2\pi i \omega \rho_0 v_0 a^3}$$

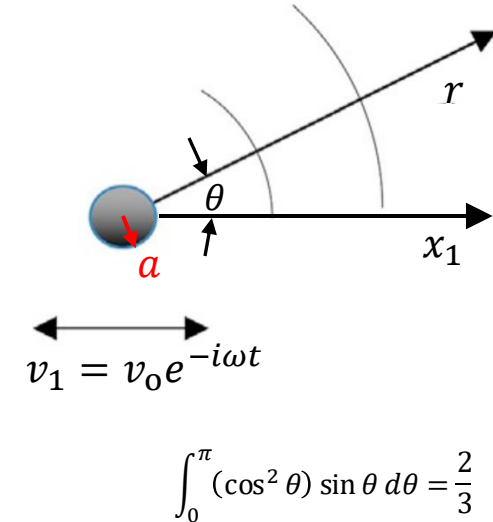
□ The acoustic field can therefore be expressed as:

$$\hat{p}(r) = \frac{ika \cos \theta}{2} \left(\frac{i\omega\rho_0 v_0 a^2 e^{ikr}}{r} \right) \left(1 - \frac{1}{ikr} \right) \Rightarrow \hat{p}(r) = -ik \cos \theta e^{ikr} \left(\frac{-2\pi i \omega \rho_0 v_0 a^3}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right)$$



$$\boxed{\hat{p}(r) = -ik \cos \theta e^{ikr} \left(\frac{\hat{F}_1}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right)}$$

➤ Later in the course, we will see that this turns out to be the **most important mechanism of sound radiation in low Mach number flows**



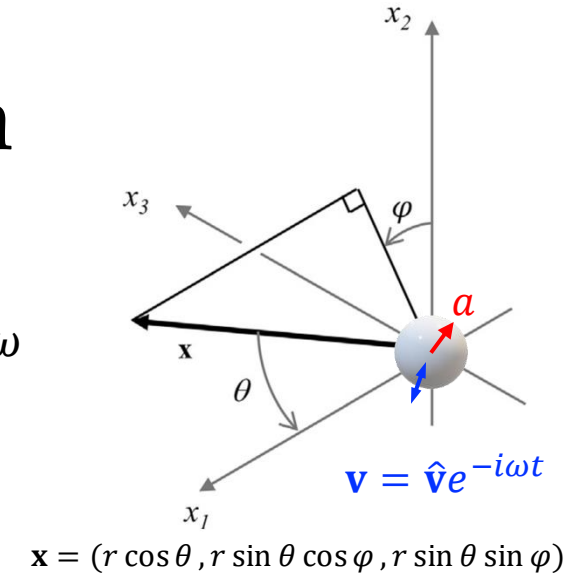
IMPORTANT RESULT

The acoustic source strength for the **transversely oscillating sphere** is determined by the force applied to the fluid

Case 3a - General spherical *surface* motion

- Consider the sound radiation from a small sphere of radius a , whose surface oscillates radially with a general surface velocity distribution $\mathbf{v}$, pulsating with ω

$$\text{BC: } \begin{cases} \mathbf{v} = \hat{\mathbf{v}}e^{-i\omega t}, & r = a \\ \frac{\partial^2 \hat{p}}{\partial x_i^2} + k^2 \hat{p} = 0, & r \geq a \end{cases}$$



- Consider the following acoustic field solution to the wave equation:

$$\hat{p}(r) = \frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{\hat{A} e^{ikr}}{r} \right) = \delta_{ij} \left[\frac{\hat{A} e^{ikr}}{r^2} \left(ik - \frac{1}{r} \right) \right] + \frac{x_i x_j}{r^2} \left[\frac{\hat{A} e^{ikr}}{r} \left(\frac{3}{r^2} - \frac{3ik}{r} - k^2 \right) \right]$$

➤ δ_{ij} – Kronecker delta function

- We can match the acoustic field to *surface motion* using the acoustic momentum equation on the surface ($r = a$)

➤ Leads to *complex expressions* that **can be simplified if limiting to acoustically compact surfaces ($ka \ll 1$)**

Case 3a - General spherical *surface* motion

- Consider the surface velocity given in spherical coordinates by:

v_o – maximum surface velocity

$$\hat{\mathbf{v}} \cdot \mathbf{n}_r = 2v_o \cos \theta \sin \theta \cos \varphi$$

- On the surface ($r = a$), we can express the surface velocity as: $[\hat{\mathbf{v}} \cdot \mathbf{n}_r]_{r=a} = 2v_o \frac{x_1 x_2}{a^2}$
- We can now match that velocity field to the pressure field expression we have with $i = 1$ and $j = 2$, yielding:

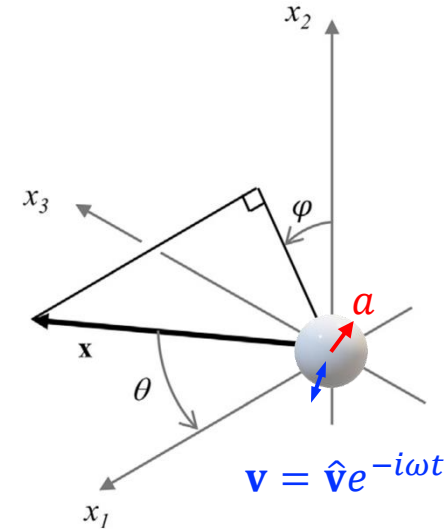
$$\hat{p}(r) = \frac{\partial^2}{\partial x_1 \partial x_2} \left(\frac{\hat{A} e^{ikr}}{r} \right) = \frac{x_1 x_2}{r^2} \left[\frac{\hat{A} e^{ikr}}{r} \left(\frac{3}{r^2} - \frac{3ik}{r} - k^2 \right) \right]$$

- As before, we solve for the constant $\hat{A}$ by matching the surface velocity to the pressure gradient in the radial direction:

$$[\hat{\mathbf{v}} \cdot \mathbf{n}_r]_{r=a} = \left[\frac{1}{i\omega \rho_o} \frac{\partial \hat{p}}{\partial r} \right]_{r=a} \Rightarrow 2v_o \frac{x_1 x_2}{a^2} = \frac{1}{i\omega \rho_o} \left[\frac{\partial \hat{p}}{\partial r} \right]_{r=a}$$

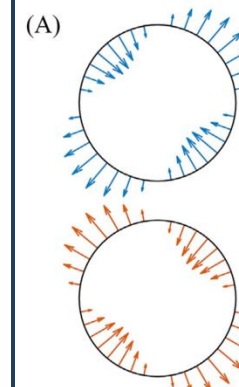
$$\Rightarrow \boxed{\begin{aligned} & 2i\omega \rho_o v_o \frac{x_1 x_2}{a^2} \\ &= \left[\frac{x_1 x_2 \hat{A} e^{ikr}}{r^2} \left\{ ik \left(\frac{3}{r^2} - \frac{3ik}{r} - k^2 \right) - \left(\frac{9}{r^3} - \frac{6ik}{r^2} - \frac{k^2}{r} \right) \right\} \right]_{r=a} \end{aligned}}$$

Aircraft Acoustics

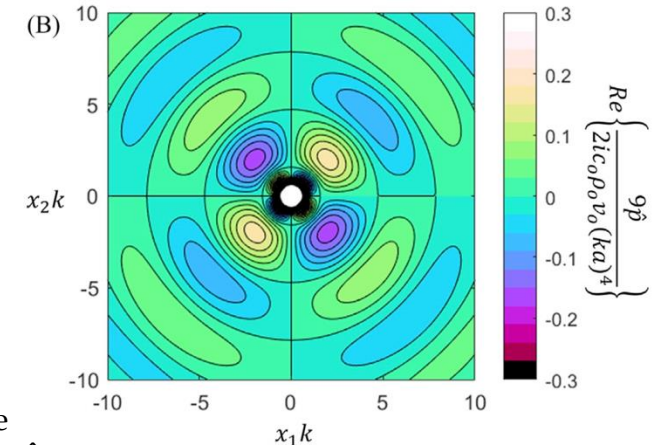


$$\mathbf{x} = (r \cos \theta, r \sin \theta \cos \varphi, r \sin \theta \sin \varphi)$$

Lateral quadrupole



Normal surface
motion amplitude $\hat{A}$
(positive & negative)
No net volume velocity



Far-field sound, $\hat{p}$

Case 3a - General spherical *surface* motion

- Limiting to an acoustically compact sphere ($ka \ll 1$) results in:

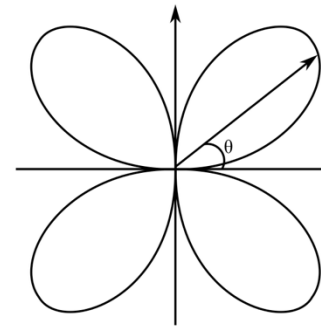
$$\hat{A} = -\frac{2i\omega\rho_0 v_0 a^4}{9}$$

$$\hat{p}(r) = \frac{2(ka)^2}{9} \left(\frac{x_1 x_2}{r^2} \right) \left(\frac{i\omega\rho_0 v_0 a^2 e^{ikr}}{r} \right) \left(1 + \frac{3i}{kr} - \frac{3}{(kr)^2} \right)$$

- Therefore, in the far-field ($kr \gg 1$) we obtain:

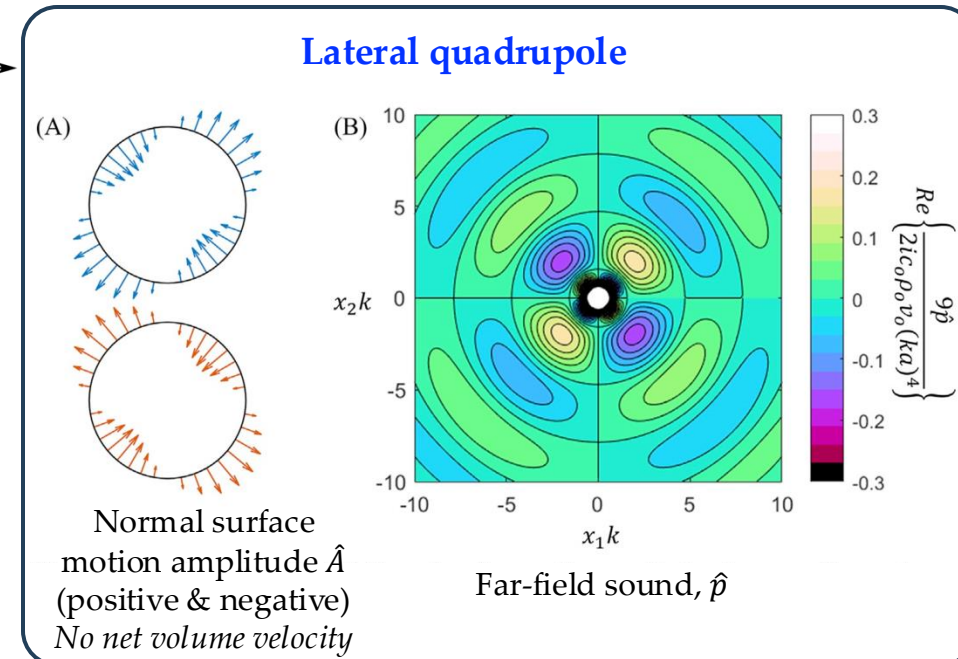
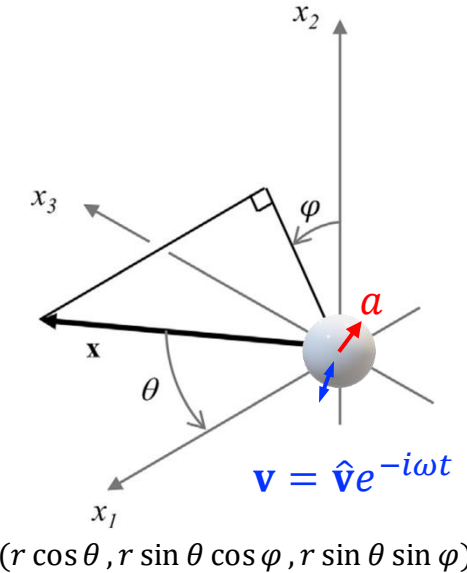
$$\hat{p}(r) = \underbrace{\frac{2(ka)^2}{9}}_{\text{Scaling factor}} \underbrace{\left(\frac{x_1 x_2}{r^2} \right)}_{\text{Directivity}} \left(\frac{i\omega\rho_0 v_0 a^2 e^{ikr}}{r} \right)$$

Scaling factor Directivity $\frac{x_1 x_2}{r^2} = \cos \theta \sin \theta \cos \varphi$



□ Lateral quadrupole

- More directional than a dipole source ($\pm 45^\circ$ beams)
- Its radiation is greatly reduced when reducing source size
 - For comparison, a dipole is proportional to $ka/2$
- No net volume flow rate ($Q=0$)**
- No net pressure applied on the fluid ($F = 0$)**



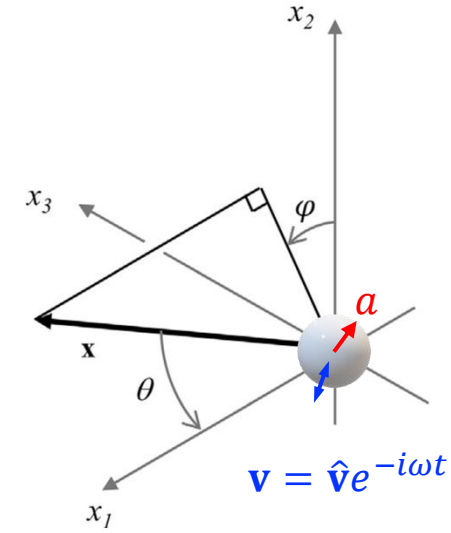
Case 3a - General spherical *surface* motion

➤ No net volume flow rate ($Q=0$)

$$Q = \int_S [\hat{\mathbf{v}} \cdot \mathbf{n}_r]_{r=a} dS = \int_0^{2\pi} \int_0^\pi 2v_o \cos \theta \sin \theta \cos \varphi e^{-i\omega t} a^2 \sin \theta d\theta d\varphi$$

$$Q = 2v_o a^2 e^{-i\omega t} \int_0^\pi \cos \theta \sin^2 \theta d\theta \int_0^{2\pi} \cos \varphi d\varphi = 0$$

$$\hat{\mathbf{v}} \cdot \mathbf{n}_r = 2v_o \cos \theta \sin \theta \cos \varphi$$



$$\mathbf{x} = (r \cos \theta, r \sin \theta \cos \varphi, r \sin \theta \sin \varphi)$$

➤ No net pressure applied on the fluid ($\mathbf{F} = 0$)

$$\hat{\mathbf{F}} = \int_S [\hat{p}]_{r=a} \mathbf{n}_r dS = 0$$

$$\hat{F}_1 = \int_0^\pi \int_0^{2\pi} \frac{2(ka)^2}{9} \cos \theta \sin \theta \cos \varphi (i\omega \rho_o v_o a e^{ika}) \cos \theta a^2 \sin \theta d\theta d\varphi$$

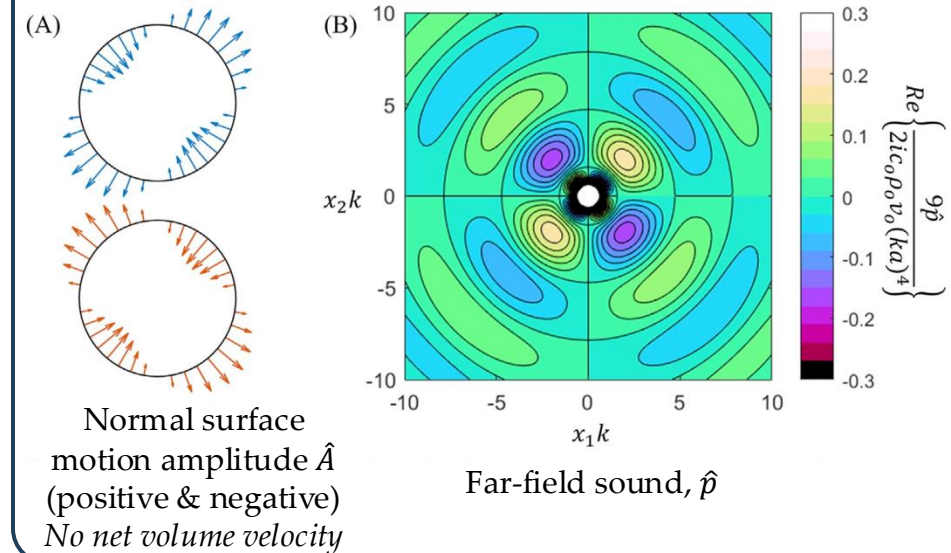
$$\hat{F}_1 = \frac{2(ka)^2}{9} (i\omega \rho_o v_o a^3 e^{ika}) \int_0^\pi \cos^2 \theta \sin^2 \theta d\theta \int_0^{2\pi} \cos \varphi d\varphi = 0$$

$$\hat{F}_2 = \frac{2(ka)^2}{9} (i\omega \rho_o v_o a^3 e^{ika}) \int_0^\pi \int_0^{2\pi} \cos \theta \sin^2 \theta \sin \theta \cos \varphi \cos \varphi d\theta d\varphi = 0$$

$$\hat{F}_3 = \frac{2(ka)^2}{9} (i\omega \rho_o v_o a^3 e^{ika}) \int_0^\pi \int_0^{2\pi} \cos \theta \sin^2 \theta \sin \theta \sin \varphi \cos \varphi d\theta d\varphi = 0$$

$$\int_0^\pi \cos \theta \sin^3 \theta d\theta = 0$$

Lateral quadrupole



Case 3b - General spherical *surface* motion

- Consider another surface velocity distribution in spherical coordinates:

v_o – maximum surface velocity

$$\hat{\mathbf{v}} \cdot \mathbf{n}_r = \frac{v_o}{2} (1 - 3 \cos^2 \theta)$$

- Consider the following acoustic field solution to the wave equation:

$$\hat{p}(r) = \frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{\hat{A} e^{ikr}}{r} \right) - \frac{k^2}{3} \left(\frac{\hat{A} e^{ikr}}{r} \right)$$

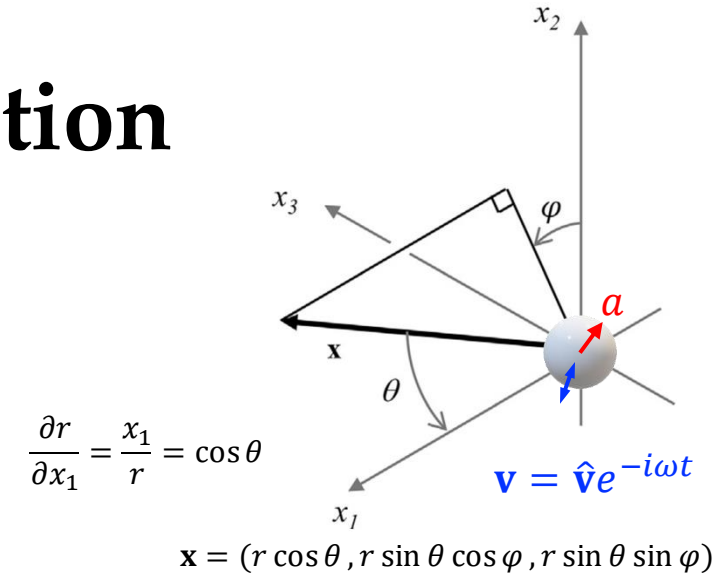
- Where: $\frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{\hat{A} e^{ikr}}{r} \right) = \delta_{ij} \left[\frac{\hat{A} e^{ikr}}{r^2} \left(ik - \frac{1}{r} \right) \right] + \frac{x_i x_j}{r^2} \left[\frac{\hat{A} e^{ikr}}{r} \left(\frac{3}{r^2} - \frac{3ik}{r} - k^2 \right) \right]$

- Matching that velocity field to the pressure field expression with $i = 1$ and $j = 1$, yields:

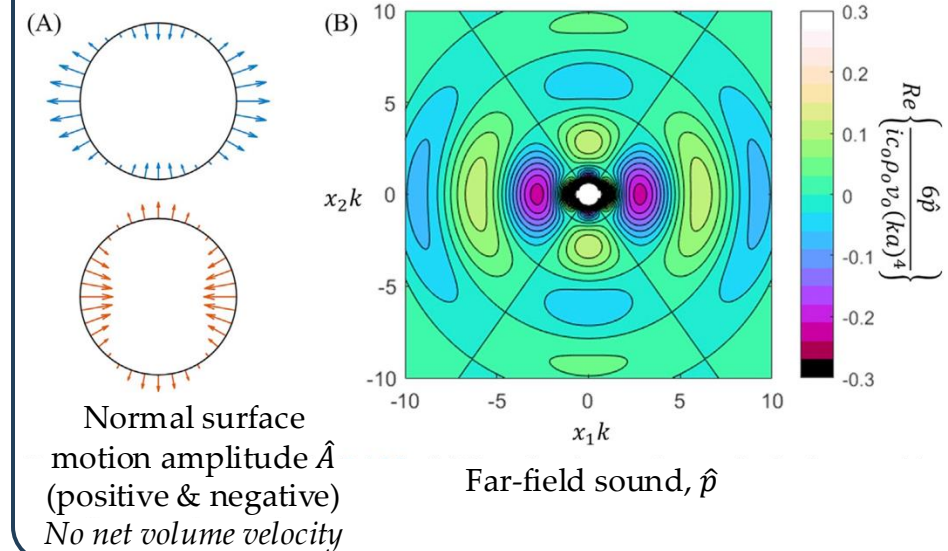
$$\hat{p}(r) = \frac{\partial^2}{\partial x_1^2} \left(\frac{\hat{A} e^{ikr}}{r} \right) - \frac{k^2}{3} \left(\frac{\hat{A} e^{ikr}}{r} \right)$$

$$\Rightarrow \hat{p}(r) = \left(\frac{\hat{A} e^{ikr}}{r} \right) (3 \cos^2 \theta - 1) \left(\frac{1}{r^2} - \frac{ik}{r} + \frac{k^2}{3} \right)$$

- where $x_1/r = \cos \theta$



longitudinal quadrupole



Case 3b - General spherical *surface* motion

- We follow the same procedure to solve the constant $\hat{A}$ by matching the pressure gradient $\partial \hat{p} / \partial r$ to the surface velocity in the radial direction:

$$[\hat{\mathbf{v}} \cdot \mathbf{n}_r]_{r=a} = \left[\frac{1}{i\omega\rho_0} \frac{\partial \hat{p}}{\partial r} \right]_{r=a} \Rightarrow \frac{v_0}{2} (1 - 3 \cos^2 \theta) = \frac{1}{i\omega\rho_0} \left[\frac{\partial \hat{p}}{\partial r} \right]_{r=a}$$

$$\hat{\mathbf{v}} \cdot \mathbf{n}_r = \frac{v_0}{2} (1 - 3 \cos^2 \theta)$$

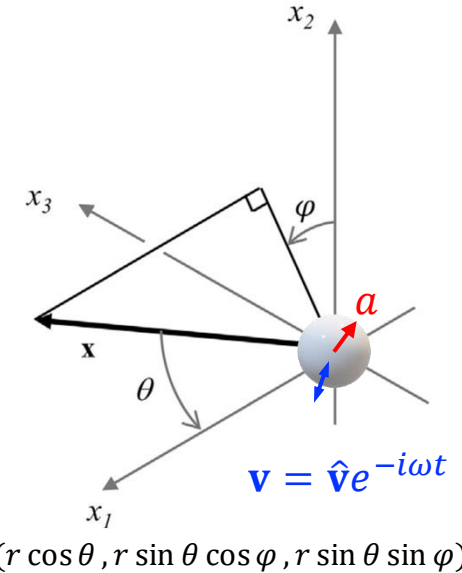
$$\frac{\partial \hat{p}}{\partial r} = (3 \cos^2 \theta - 1) \left(\frac{\hat{A} e^{ikr}}{r} \right) \left\{ ik \left(\frac{1}{r^2} - \frac{ik}{r} + \frac{k^2}{3} \right) - \left(\frac{3}{r^3} - \frac{2ik}{r^2} + \frac{k^2}{3r} \right) \right\}$$

$$\Rightarrow \frac{v_0}{2} (1 - 3 \cos^2 \theta) = \left[(3 \cos^2 \theta - 1) \left(\frac{\hat{A} e^{ikr}}{i\omega\rho_0 r} \right) \left\{ ik \left(\frac{1}{r^2} - \frac{ik}{r} + \frac{k^2}{3} \right) - \left(\frac{3}{r^3} - \frac{2ik}{r^2} + \frac{k^2}{3r} \right) \right\} \right]_{r=a}$$

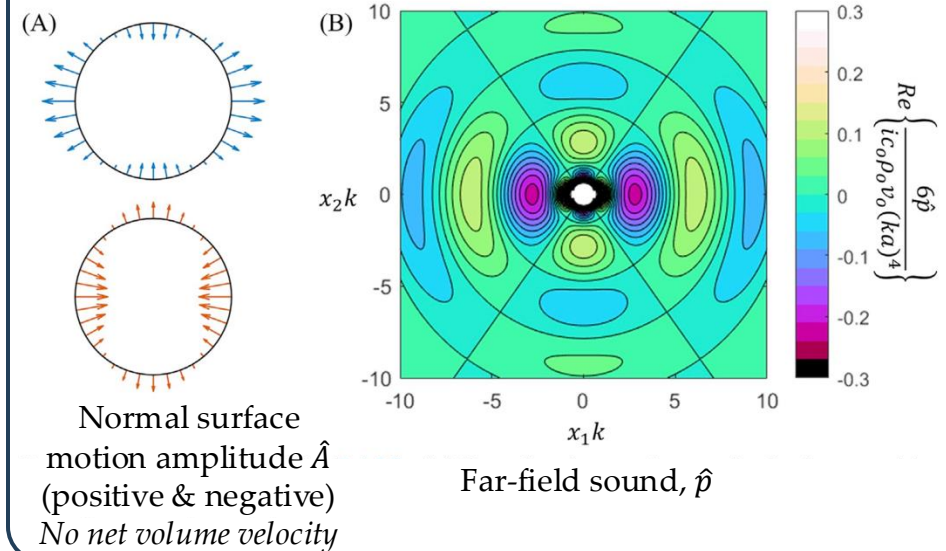
$$\Rightarrow \hat{A} = \frac{1}{2} \left(\frac{i\omega\rho_0 v_0 a^4 e^{-ika}}{3 \left(1 - \frac{2ika}{3} + \frac{(ka)^2}{9} \right) - ika \left(1 - ika + \frac{(ka)^2}{3} \right)} \right)$$

- Limiting to an acoustically compact sphere ($ka \ll 1$) results in:

$$\hat{A} = \frac{i\omega\rho_0 v_0 a^4}{6}$$



longitudinal quadrupole



Case 3b - General spherical *surface* motion

$$\hat{p}(r) = \frac{(ka)^2}{6} \frac{i\omega\rho_0 v_0 a^2 e^{ikr}}{r} (3 \cos^2 \theta - 1) \left(\frac{1}{(kr)^2} - \frac{i}{kr} + \frac{1}{3} \right)$$

- Therefore, in the far-field ($kr \gg 1$) we obtain:

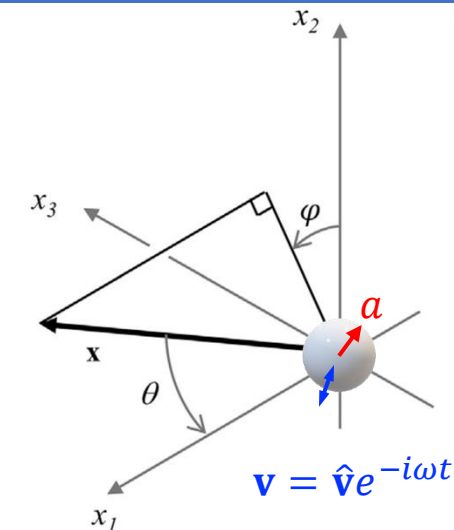
$$\hat{p}(r) = \underbrace{\frac{(ka)^2}{6}}_{\text{Scaling factor}} \underbrace{\left(\cos^2 \theta - \frac{1}{3} \right)}_{\text{Directivity}} \left(\frac{i\omega\rho_0 v_0 a^2 e^{ikr}}{r} \right)$$

Scaling factor Directivity

□ Longitudinal quadrupole

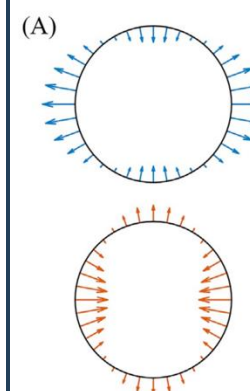
- Radiation is further reduced when reducing source size
- A dipole is proportional to $ka/2$
 - A lateral quadrupole is proportional to $2ka/9$
- Similar to a lateral quadrupole:
- **No net volume flow rate ($Q=0$)**
 - **No net pressure applied on the fluid ($F = 0$)**

$(Ka)^2$

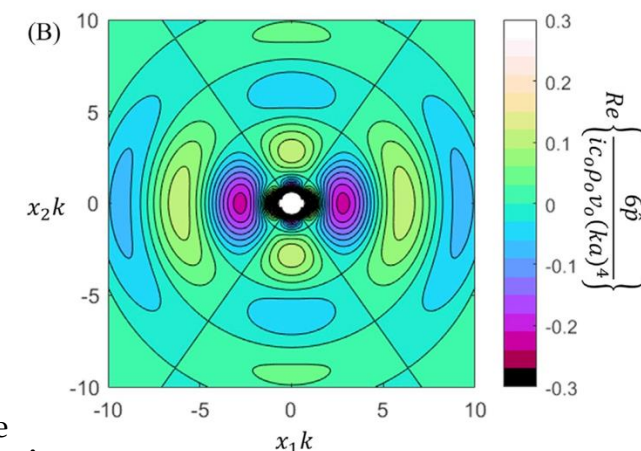


$$\mathbf{x} = (r \cos \theta, r \sin \theta \cos \phi, r \sin \theta \sin \phi)$$

longitudinal quadrupole



Normal surface
motion amplitude $\hat{A}$
(positive & negative)
No net volume velocity



Far-field sound, $\hat{p}$

Case 3b - General spherical *surface* motion

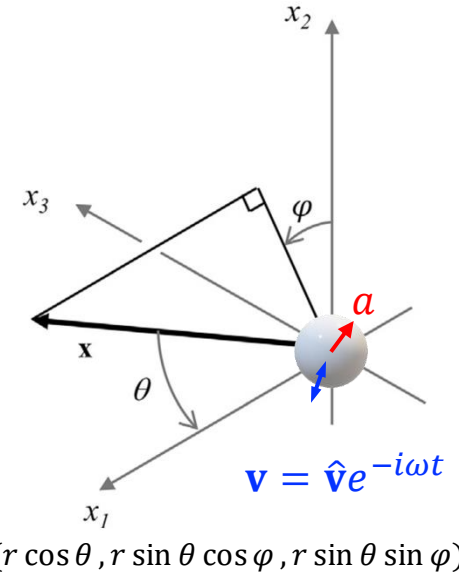
➤ No net volume flow rate ($Q=0$)

$$Q = \int_S [\hat{\mathbf{v}} \cdot \mathbf{n}_r]_{r=a} dS = \frac{v_0}{2} \int_0^{2\pi} \int_0^\pi (1 - 3 \cos^2 \theta) e^{-i\omega t} a^2 \sin \theta d\theta d\varphi$$

$$Q = v_0 \pi a^2 e^{-i\omega t} \int_0^\pi (1 - 3 \cos^2 \theta) \sin \theta d\theta = 0$$

$$\hat{p}(r) = \frac{(ka)^2}{6} \left(\cos^2 \theta - \frac{1}{3} \right) \left(\frac{i\omega \rho_0 v_0 a^2 e^{ikr}}{r} \right)$$

$$\hat{\mathbf{v}} \cdot \mathbf{n}_r = \frac{v_0}{2} (1 - 3 \cos^2 \theta)$$



➤ No net pressure applied on the fluid ($\mathbf{F} = 0$)

$$\hat{\mathbf{F}} = \int_S [\hat{p}]_{r=a} \mathbf{n}_r dS = 0$$

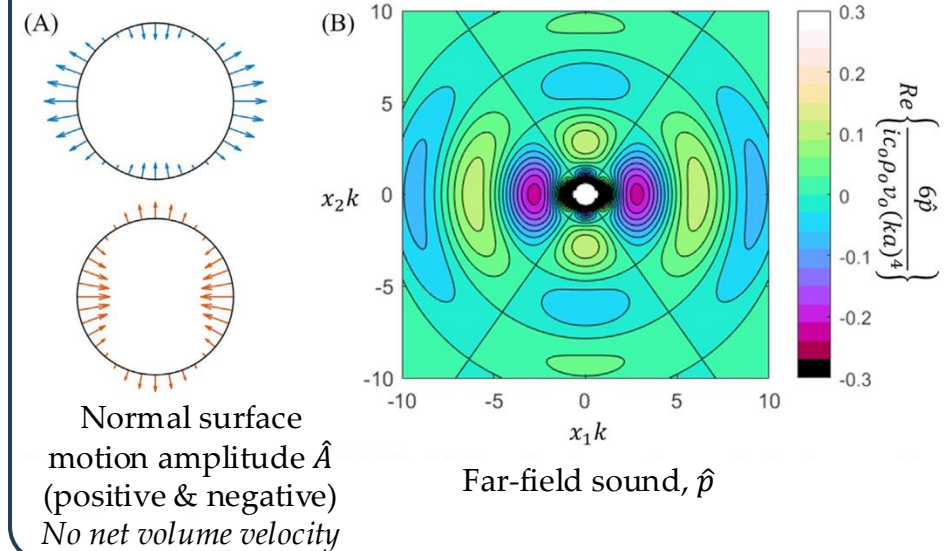
$$\hat{F}_1 = \int_0^\pi \int_0^{2\pi} \frac{(ka)^2}{6} \left(\cos^2 \theta - \frac{1}{3} \right) (i\omega \rho_0 v_0 a e^{ika}) \cos \theta a^2 \sin \theta d\theta d\varphi$$

$$\hat{F}_1 = \frac{(ka)^2}{3} (i\pi \omega \rho_0 v_0 a^3 e^{ika}) \int_0^\pi \left(\cos^3 \theta \sin \theta - \frac{1}{3} \cos \theta \sin \theta \right) d\theta = 0$$

$$\hat{F}_2 = \frac{(ka)^2}{3} (i\omega \rho_0 v_0 a^3 e^{ika}) \int_0^\pi \left(\cos^2 \theta \sin \theta - \frac{1}{3} \sin \theta \right) \sin \theta d\theta \int_0^{2\pi} \cos \varphi d\varphi = 0$$

$$\hat{F}_3 = \frac{(ka)^2}{3} (i\omega \rho_0 v_0 a^3 e^{ika}) \int_0^\pi \left(\cos^2 \theta \sin \theta - \frac{1}{3} \sin \theta \right) \sin \theta d\theta \int_0^{2\pi} \sin \varphi d\varphi = 0$$

longitudinal quadrupole

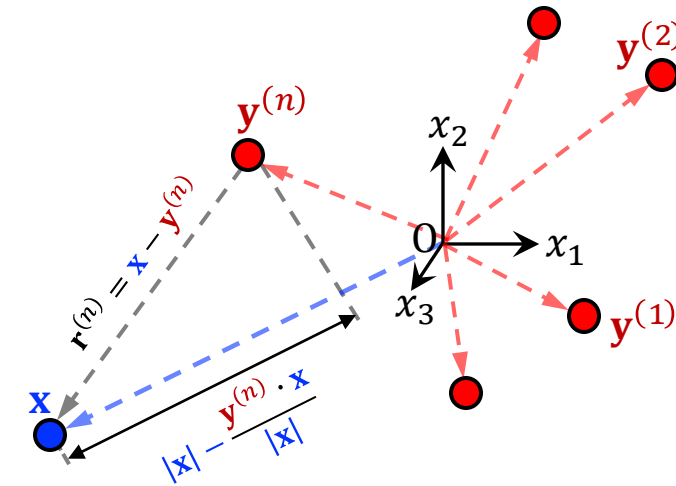


Superposition of acoustic sources

- ❑ The linearity of the wave equation suggests that any number of sound fields can be added together
- ❑ Consider a region containing a N number of monopole sources located at $\mathbf{y}^{(n)}$, where $n = 1, 2, 3, \dots, N$
- ❑ The acoustic field is evaluated at point $\mathbf{x}$ (observer location) by summing the individual acoustic source fields:

$$\hat{p}(\mathbf{x}) = \sum_{n=1}^N \frac{\hat{A}_n e^{ik|\mathbf{x}-\mathbf{y}^{(n)}|}}{|\mathbf{x}-\mathbf{y}^{(n)}|}$$

- $r^{(n)} = |\mathbf{x} - \mathbf{y}^{(n)}| \rightarrow$ distance from a source located at $\mathbf{y}^{(n)}$ to observer location $\mathbf{x}$
- $\hat{A}_n \rightarrow$ relative amplitude contribution of each source
 - For a pulsating sphere – $\hat{A}_n = -i\omega\rho_0\hat{Q}/4\pi$



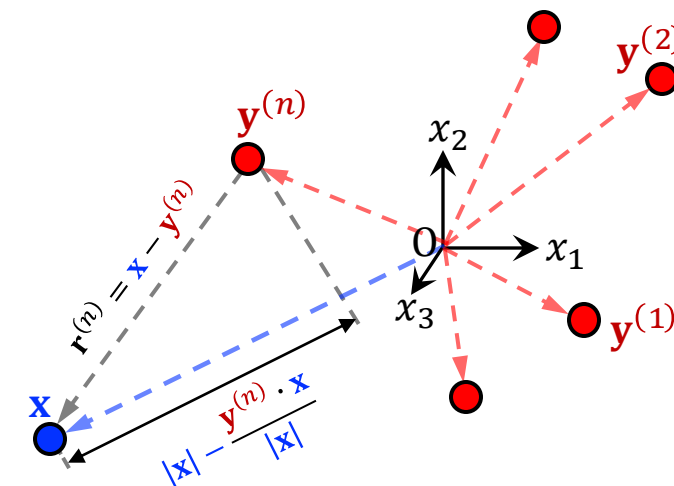
Superposition of acoustic sources

- In the **geometric far-field**, $|\mathbf{x}| \gg |\mathbf{y}|$
 - We chose the coordinate origin to be located near the sources
 - Thus, $r^{(n)}$ can be approximated (up to a first-order) using a Taylor series expansion:

$$r(x_i, y_i) \approx \underbrace{r(x_i, 0)}_{r(x_i, 0) = |\mathbf{x}|} - y_i \underbrace{\frac{\partial r(x_i, 0)}{\partial x_i}}_{\frac{\partial r(x_i, 0)}{\partial x_i} = \frac{x_i}{|\mathbf{x}|}} + \frac{y_i y_j}{2} \frac{\partial^2 r(x_i, 0)}{\partial x_i \partial x_j} + \dots \Rightarrow \boxed{r(x_i, y_i) = |\mathbf{x} - \mathbf{y}| \approx |\mathbf{x}| - \frac{\mathbf{y} \cdot \mathbf{x}}{|\mathbf{x}|}}$$

Keeping 1st-order terms

$r(x_i, y_i) = |\mathbf{x} - \mathbf{y}| = \sqrt{(x_i - y_i)^2}$



- Substituting into the acoustic pressure field results in: $\hat{p}(\mathbf{x}) \approx \frac{e^{ik|\mathbf{x}|}}{|\mathbf{x}|} \sum_{n=1}^N \frac{\hat{A}_n e^{-iky^{(n)} \cdot \mathbf{x}/|\mathbf{x}|}}{1 - \mathbf{y}^{(n)} \cdot \mathbf{x}/|\mathbf{x}|^2} \quad |\mathbf{x}| \gg |\mathbf{y}^{(n)}|$
- The term in the denominator can be further approximated using a Laurent series around $1/|\mathbf{x}|^2 = \varepsilon \rightarrow 0$ to yield:

Laurent series around $\varepsilon \rightarrow 0$:

$$\frac{1}{1 - A\varepsilon} = 1 + A\varepsilon - A^2\varepsilon^2 + \dots$$



$$\boxed{\hat{p}(\mathbf{x}) \approx \frac{e^{ik|\mathbf{x}|}}{|\mathbf{x}|} \sum_{n=1}^N \hat{A}_n e^{-iky^{(n)} \cdot \mathbf{x}/|\mathbf{x}|}} \quad |\mathbf{x}| \gg |\mathbf{y}^{(n)}|$$

Dipole - clustering two monopoles

- Consider two monopole sources of equal strength and opposite phase placed at a **small distance d** apart to form a **compact pair ($kd \ll 1$)**
 - *Negative (-)* source is placed at $\mathbf{y}^{(1)} = (-d/2, 0, 0)$, with $\hat{A}_1 = -i\omega\rho_0\hat{Q}/4\pi$
 - *Positive (+)* source is placed at $\mathbf{y}^{(2)} = (+d/2, 0, 0)$, with $\hat{A}_2 = i\omega\rho_0\hat{Q}/4\pi$
- The acoustic far-field field is evaluated at point $\mathbf{x}$ (observer location) by summing the contributions from the two acoustic source fields:

$$\hat{p}(\mathbf{x}) = \frac{i\omega\rho_0\hat{Q}}{4\pi} \left(\frac{e^{ikr^{(2)}}}{r^{(2)}} - \frac{e^{ikr^{(1)}}}{r^{(1)}} \right)$$

- $r^{(n)} = |\mathbf{x} - \mathbf{y}^{(n)}| \rightarrow$ distance from a source located at $\mathbf{y}^{(n)}$ to observer location $\mathbf{x}$
- **$d/2$ is small**, and so we can use a Taylor series expansion for $e^{ikr^{(1)}}/r^{(1)}$ & $e^{ikr^{(2)}}/r^{(2)}$ about the origin $(0,0,0)$:

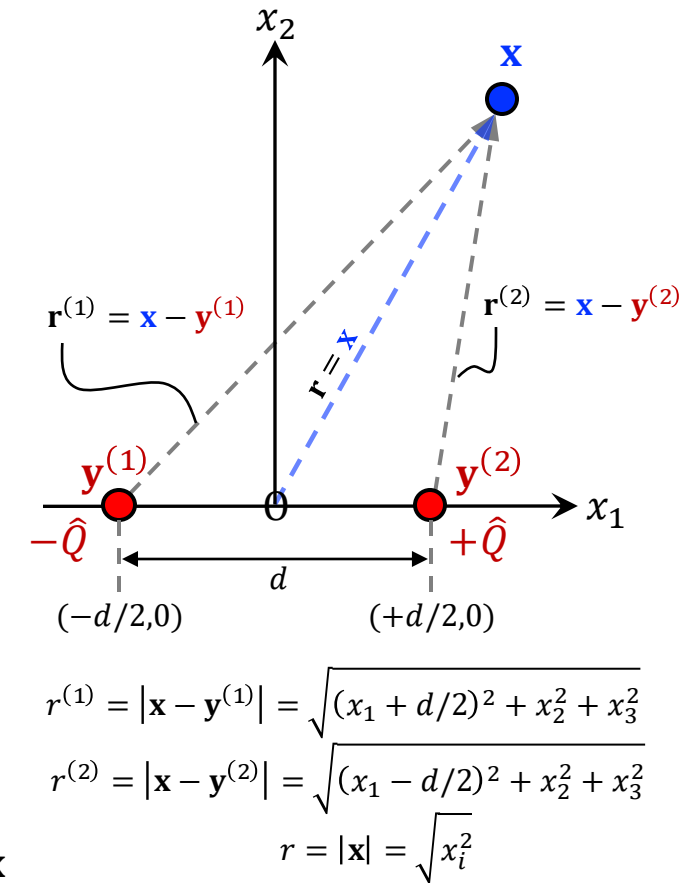
$$\frac{e^{ikr^{(1)}}}{r^{(1)}} = \frac{e^{ikr}}{r} - \frac{d}{2} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3)$$

$$\frac{e^{ikr^{(2)}}}{r^{(2)}} = \frac{e^{ikr}}{r} + \frac{d}{2} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3)$$

$r \gg d$

$$\hat{p}(\mathbf{x}) \approx \frac{i\omega\rho_0\hat{Q}d}{4\pi} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right)$$

A familiar term we solved before...



Dipole - clustering two monopoles

$$\hat{p}(\mathbf{x}) \approx \frac{i\omega\rho_0\hat{Q}d}{4\pi} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) = \frac{i\omega\rho_0\hat{Q}d}{4\pi} \frac{\partial r}{\partial x_1} \frac{\partial}{\partial r} \left(\frac{e^{ikr}}{r} \right)$$

$$\hat{p}(\mathbf{x}) \approx ikd \cos \theta \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right) \quad \begin{matrix} r \gg d \\ kd \ll 1 \end{matrix}$$

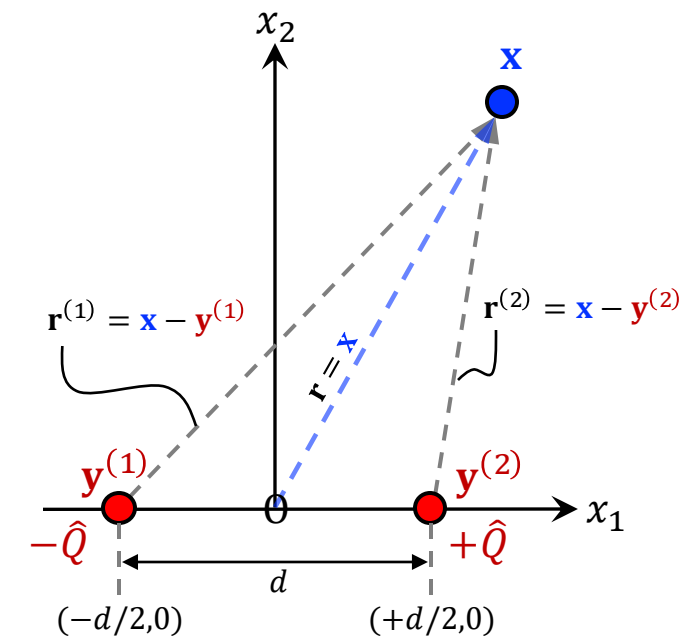
□ Similar to the result of a **translating sphere** with radius a :

$$\hat{Q} = v_0 4\pi a^2 \quad \hat{p}(r) = \underbrace{\frac{ika}{2}}_{\text{Scaling factor}} \underbrace{\cos \theta}_{\text{Directivity}} \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right)$$

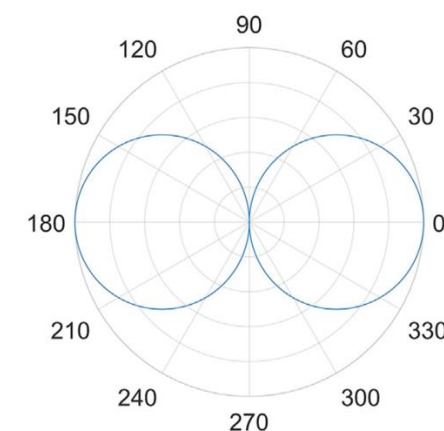
Scaling factor Directivity

□ **Acoustic near-field, $kr \ll 1$** → $\left(\frac{d}{r^2} \right)$ scaling $\hat{p}(\mathbf{x}) = \cos \theta \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi} \right) \left(\frac{d}{r^2} \right)$

□ **Acoustic far-field, $kr \gg 1$** → $\left(\frac{kd}{r} \right)$ scaling $\hat{p}(\mathbf{x}) = \cos \theta \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi} \right) \left(\frac{kd}{r} \right)$



$$\frac{\partial r}{\partial x_1} = \frac{x_1}{r} = \cos \theta \quad r = |\mathbf{x}| = \sqrt{x_i^2}$$



Quadrupole - clustering two dipoles

- ❑ Consider two dipole sources of equal strength and opposite phase placed at a **small distance d** apart to form a **compact pair ($kd \ll 1$)**
 - *Negative (-)* dipole is placed at $\mathbf{y}^{(1)} = (-d/2, 0, 0)$
 - *Positive (+)* dipole is placed at $\mathbf{y}^{(2)} = (+d/2, 0, 0)$
- ❑ The acoustic far-field field is evaluated at point $\mathbf{x}$ (observer location) by summing the contributions from the two acoustic dipole fields:

$$\hat{p}(\mathbf{x}) \approx \frac{i\omega\rho_0\hat{Q}d}{4\pi} \left[\frac{\partial}{\partial x_1} \left(\frac{e^{ikr^{(2)}}}{r^{(2)}} \right) - \frac{\partial}{\partial x_1} \left(\frac{e^{ikr^{(1)}}}{r^{(1)}} \right) \right]$$

- $r^{(n)} = |\mathbf{x} - \mathbf{y}^{(n)}| \rightarrow$ distance from a dipole located at $\mathbf{y}^{(n)}$ to observer location $\mathbf{x}$
- **$d/2$ is small**, and so we can use a Taylor series expansion as before about the origin $(0,0,0)$:

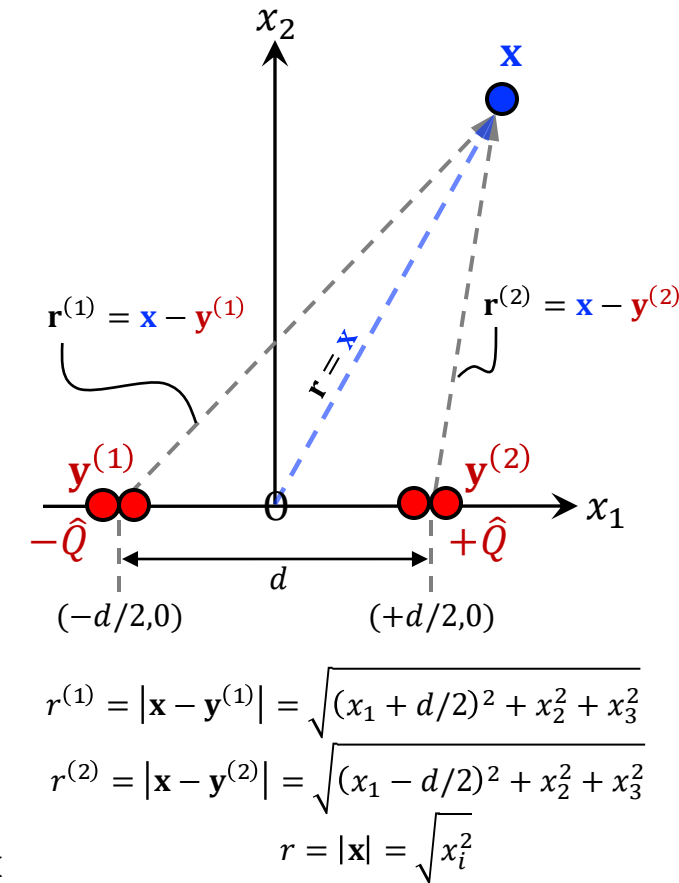
$$\begin{aligned} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr^{(1)}}}{r^{(1)}} \right) &= \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) - \frac{d}{2} \frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \frac{\partial^3}{\partial x_1^3} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3) \\ \frac{\partial}{\partial x_1} \left(\frac{e^{ikr^{(2)}}}{r^{(2)}} \right) &= \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) + \frac{d}{2} \frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \frac{\partial^3}{\partial x_1^3} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3) \end{aligned}$$

$r \gg d$

$$\hat{p}(\mathbf{x}) \approx \frac{i\omega\rho_0\hat{Q}d^2}{4\pi} \frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right)$$

A familiar term we solved before...

$$\frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{e^{ikr}}{r} \right) = \delta_{ij} \left[\frac{e^{ikr}}{r^2} \left(ik - \frac{1}{r} \right) \right] + \frac{x_i x_j}{r^2} \left[\frac{e^{ikr}}{r} \left(\frac{3}{r^2} - \frac{3ik}{r} - k^2 \right) \right]$$



Quadrupole - clustering two dipoles

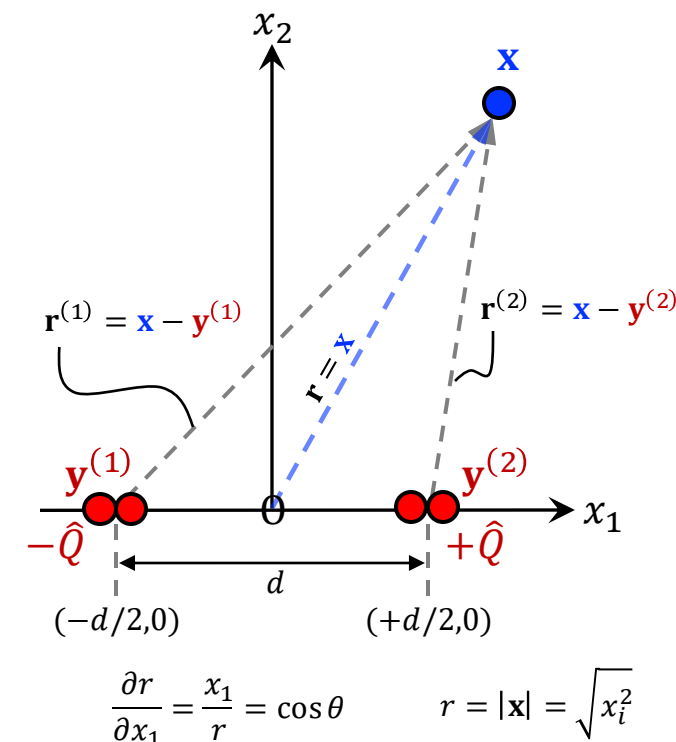
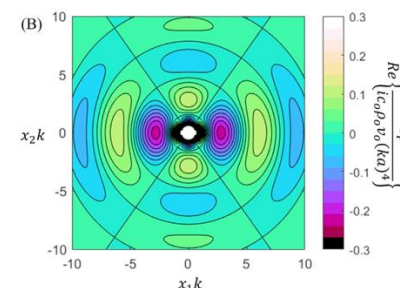
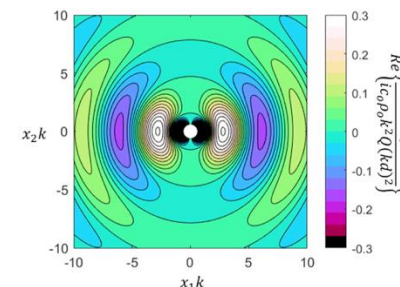
$$\hat{p}(\mathbf{x}) \approx \frac{i\omega\rho_0\hat{Q}d^2}{4\pi} \frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) = \frac{i\omega\rho_0\hat{Q}d^2}{4\pi} \left(\frac{k^2 e^{ikr}}{r} \right) \left[\left(\frac{i}{kr} - \frac{1}{(kr)^2} \right) + 3 \left(\frac{x_1}{r} \right)^2 \left(\frac{1}{(kr)^2} - \frac{i}{kr} - \frac{1}{3} \right) \right]$$

□ Acoustic far-field, $kr \gg 1 \rightarrow (kd)^2$ scaling

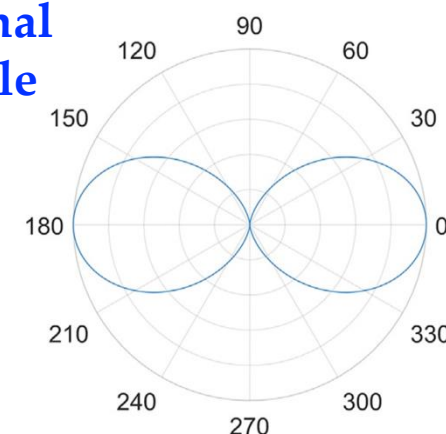
$$\hat{p}(\mathbf{x}) \approx (kd)^2 \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi r} \right) \cos^2 \theta \quad \begin{matrix} r \gg d \\ kd \ll 1 \end{matrix}$$

□ Almost Similar to the results of a sphere of radius a with a general spherical surface motion (Case 3b):

$$\hat{Q} = v_o 4\pi a^2 \quad \hat{p}(r) = \underbrace{\frac{(ka)^2}{6}}_{\text{Scaling factor}} \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi r} \right) \underbrace{\left(\cos^2 \theta - \frac{1}{3} \right)}_{\text{Directivity}}$$



Longitudinal quadrupole



Quadrupole - clustering four monopoles

- ❑ Consider the following arrangement of **four monopole sources**:
 - Two *negative* (-) sources placed at $\mathbf{y}^{(1)} = (d/2, d/2, 0)$ and $\mathbf{y}^{(2)} = (-d/2, -d/2, 0)$
 - Two *positive* (+) sources placed at $\mathbf{y}^{(3)} = (d/2, -d/2, 0)$ and $\mathbf{y}^{(4)} = (-d/2, d/2, 0)$
 - Assume a **small distance** d to form a **compact arrangement** ($kd \ll 1$)
- ❑ The acoustic far-field field is evaluated at point $\mathbf{x}$ (observer location) by summing the contributions from the four acoustic source fields:

$$\hat{p}(\mathbf{x}) = \frac{i\omega\rho_0\hat{Q}}{4\pi} \left(\frac{e^{ikr^{(3)}}}{r^{(3)}} - \frac{e^{ikr^{(1)}}}{r^{(1)}} + \frac{e^{ikr^{(4)}}}{r^{(4)}} - \frac{e^{ikr^{(2)}}}{r^{(2)}} \right)$$

- $d/2$ is small $\rightarrow$ Taylor series expansion as before about the origin (0,0,0):

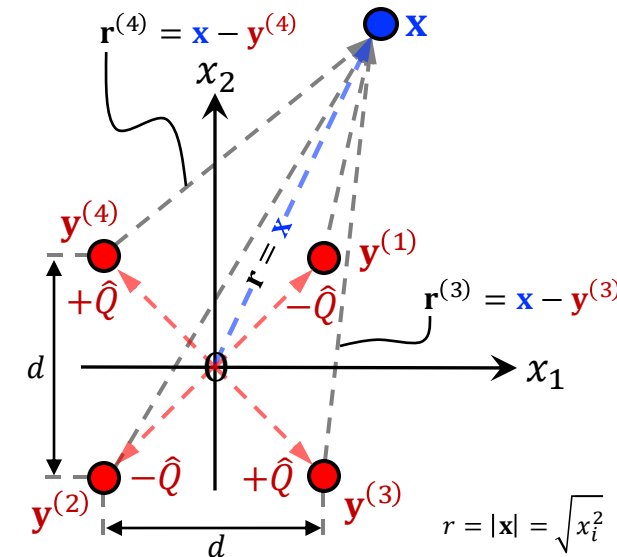
$$\begin{aligned} \frac{e^{ikr^{(1)}}}{r^{(1)}} &= \frac{e^{ikr}}{r} + \frac{d}{2} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \left[\frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \frac{\partial^2}{\partial x_2^2} \left(\frac{e^{ikr}}{r} \right) \right] + \frac{2}{2!} \frac{d^2}{4} \frac{\partial^2}{\partial x_1 \partial x_2} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3) \\ \frac{e^{ikr^{(2)}}}{r^{(2)}} &= \frac{e^{ikr}}{r} - \frac{d}{2} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \left[\frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \frac{\partial^2}{\partial x_2^2} \left(\frac{e^{ikr}}{r} \right) \right] + \frac{2}{2!} \frac{d^2}{4} \frac{\partial^2}{\partial x_1 \partial x_2} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3) \\ \frac{e^{ikr^{(3)}}}{r^{(3)}} &= \frac{e^{ikr}}{r} + \frac{d}{2} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \left[\frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \frac{\partial^2}{\partial x_2^2} \left(\frac{e^{ikr}}{r} \right) \right] - \frac{2}{2!} \frac{d^2}{4} \frac{\partial^2}{\partial x_1 \partial x_2} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3) \\ \frac{e^{ikr^{(4)}}}{r^{(4)}} &= \frac{e^{ikr}}{r} - \frac{d}{2} \frac{\partial}{\partial x_1} \left(\frac{e^{ikr}}{r} \right) + \frac{1}{2!} \frac{d^2}{4} \left[\frac{\partial^2}{\partial x_1^2} \left(\frac{e^{ikr}}{r} \right) + \frac{\partial^2}{\partial x_2^2} \left(\frac{e^{ikr}}{r} \right) \right] - \frac{2}{2!} \frac{d^2}{4} \frac{\partial^2}{\partial x_1 \partial x_2} \left(\frac{e^{ikr}}{r} \right) + \dots + O(d^3) \end{aligned}$$

 $r \gg d$

$$\hat{p}(\mathbf{x}) \approx -\frac{i\omega\rho_0\hat{Q}d^2}{4\pi} \frac{\partial^2}{\partial x_1 \partial x_2} \left(\frac{e^{ikr}}{r} \right)$$

A familiar term we solved before...

$$\frac{\partial^2}{\partial x_i \partial x_j} \left(\frac{e^{ikr}}{r} \right) = \delta_{ij} \left[\frac{e^{ikr}}{r^2} \left(ik - \frac{1}{r} \right) \right] + \frac{x_i x_j}{r^2} \left[\frac{e^{ikr}}{r} \left(\frac{3}{r^2} - \frac{3ik}{r} - k^2 \right) \right]$$



$$\begin{aligned} r^{(1)} &= |\mathbf{x} - \mathbf{y}^{(1)}| = \sqrt{(x_1 - d/2)^2 + (x_2 - d/2)^2 + x_3^2} \\ r^{(2)} &= |\mathbf{x} - \mathbf{y}^{(2)}| = \sqrt{(x_1 + d/2)^2 + (x_2 + d/2)^2 + x_3^2} \\ r^{(3)} &= |\mathbf{x} - \mathbf{y}^{(3)}| = \sqrt{(x_1 - d/2)^2 + (x_2 + d/2)^2 + x_3^2} \\ r^{(4)} &= |\mathbf{x} - \mathbf{y}^{(4)}| = \sqrt{(x_1 + d/2)^2 + (x_2 - d/2)^2 + x_3^2} \end{aligned}$$

Quadrupole - clustering four monopoles

$$\hat{p}(\mathbf{x}) \approx -\frac{i\omega\rho_0\hat{Q}d^2}{4\pi}\frac{\partial^2}{\partial x_1\partial x_2}\left(\frac{e^{ikr}}{r}\right) = k^2\frac{i\omega\rho_0\hat{Q}d^2}{4\pi}\frac{x_1x_2}{r^2}\left[\frac{e^{ikr}}{r}\left(1 + \frac{3i}{kr} - \frac{3}{(kr)^2}\right)\right]$$

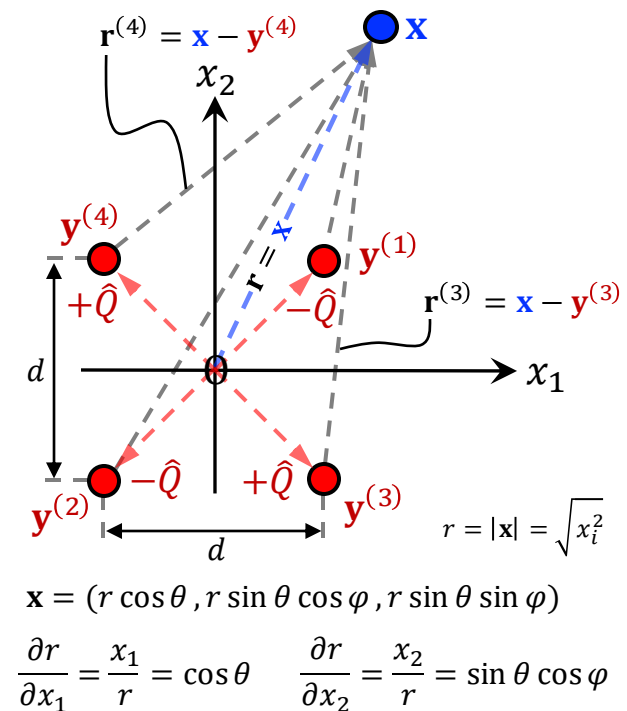
□ **Acoustic far-field, $kr \gg 1$** $\rightarrow (kd)^2$ scaling

$$\hat{p}(\mathbf{x}) \approx (kd)^2 \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi r} \right) \left(\frac{x_1x_2}{r^2} \right) \quad \begin{matrix} r \gg d \\ kd \ll 1 \end{matrix}$$

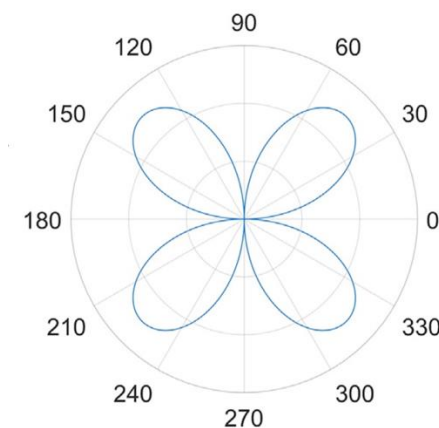
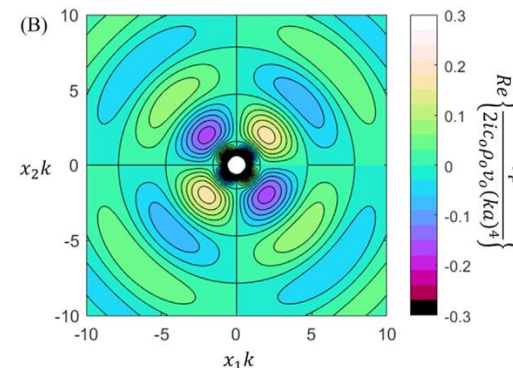
□ *Almost Similar* to the results of a sphere of radius a with a **general spherical surface motion (Case 3b)**:

$$\hat{Q} = v_o 4\pi a^2 \quad \hat{p}(r) = \underbrace{\frac{2(ka)^2}{9}}_{\text{Scaling factor}} \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi r} \right) \underbrace{\left(\frac{x_1x_2}{r^2} \right)}_{\text{Directivity}}$$

$$\left(\frac{x_1x_2}{r^2} \right) = \cos\theta \sin\theta \cos\varphi = \frac{1}{2} \sin 2\theta \cos\varphi$$



Lateral quadrupole



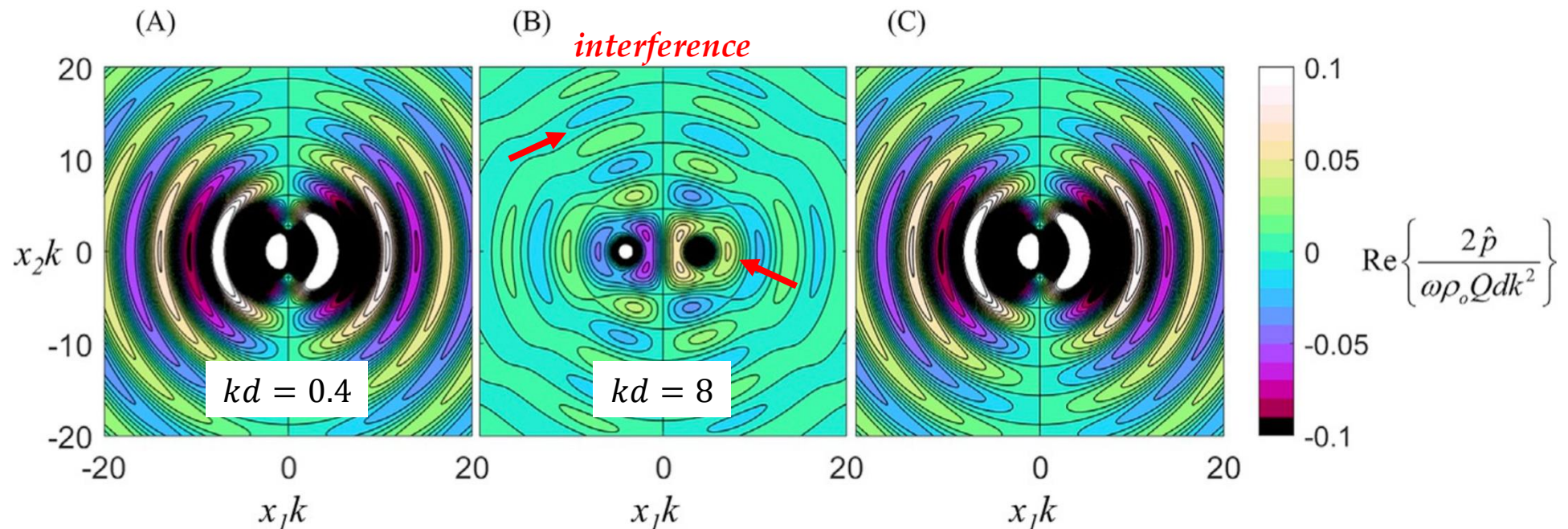
Compact vs. non-compact source

□ (A) & (B)

- *Exact* sound field radiated by **two monopoles of opposite phase** spaced at *compact* ($kd \ll 1$) and *non-compact* distances

□ (C)

- *Approximate* dipole sound field by: $\hat{p}(\mathbf{x}) \approx ikd \cos \theta \left(\frac{i\omega\rho_0\hat{Q}e^{ikr}}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right)$



Acoustic intensity

- Recall the **acoustic sound power output** W_a of a source within a CV defined by the surface S :

$$W_a = \int_S \mathbf{I} \cdot \mathbf{n}^{(o)} dS$$

- Where for a **stationary ($\mathbf{U} = 0$) homentropic medium**, the **acoustic intensity** $\mathbf{I}$ is written as:

➤ Recall $\mathbf{v} = \mathbf{v}'$

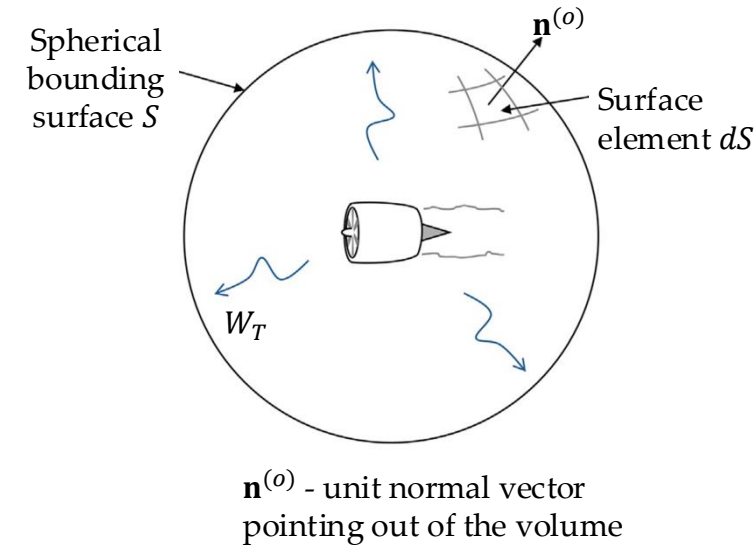
$$\mathbf{I} = E[p'\mathbf{v}']$$

- For **linear acoustics and harmonic time dependence**:

$$\mathbf{I}_j = E[\underbrace{|\hat{p}| \cos(\omega t + \phi_p)}_{\text{Complex pressure}} \underbrace{|\hat{v}_j| \cos(\omega t + \phi_j)}_{\text{Complex velocity component}}]$$

Complex
pressure

Complex velocity
component



E – **expected value** – mean value of a stochastic variable taken over many repeated realizations N (under identical conditions)

$$E[p'] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N p'_n(t)$$

$p'_n(t)$ – n -th sample of p'
 N – total samples taken

Acoustic intensity

- Using trigonometry, we can write:

$$\mathbf{I}_j = E[|\hat{p}| \cos(\omega t + \phi_p) |\hat{v}_j| \cos(\omega t + \phi_j)] \Rightarrow \mathbf{I}_j = \frac{|\hat{p}| |\hat{v}_j|}{2} E[\cos(2\omega t + \phi_p + \phi_j) + \cos(\phi_p - \phi_j)]$$

$\cos(\alpha + \beta) \cos(\alpha + \gamma) = \frac{1}{2} \cos(2\alpha + \beta + \gamma) + \cos(\beta - \gamma)$ 

- Expected value of a harmonic time history is zero! $E[\cos(2\omega t + \phi_p + \phi_j)] \equiv 0$
- Therefore, for a **harmonic time signal**, we obtain:

$$\mathbf{I}_j = \frac{|\hat{p}| |\hat{v}_j|}{2} \cos(\phi_p - \phi_j) = \frac{\Re(\hat{p} \hat{v}_j^*)}{2}$$

- $\hat{v}_j^*$ - complex conjugate of the velocity of a harmonic wave

Acoustic intensity

- ❑ Consider a **spherical surface** $|\mathbf{x}| = \text{const.}$ that **encloses acoustic sources** and lies in the **acoustic far-field**
- ❑ The equation we derived shows that the acoustic intensity component $\mathbf{I}_r$ will depend on the particle velocity in the radial direction $\hat{v}_r$
- ❑ And the momentum conservation shows that: $\hat{\mathbf{v}} = \frac{1}{i\omega\rho_o} \nabla \hat{p}$
 - $\hat{v}_r$ is dependent on $\partial \hat{p} / \partial r$
- ❑ Recall, that the **far-field sound field** for all source distributions has a **variation in the radial direction** given by $\hat{p} \propto e^{ikr}/r$, where $r = |\mathbf{x}|$

$$\hat{v}_r = \frac{1}{i\omega\rho_o} \frac{\partial \hat{p}}{\partial r} \Rightarrow \hat{v}_r = \frac{1}{i\omega\rho_o} \frac{\partial}{\partial r} \left(\frac{e^{ikr}}{r} \right) \Rightarrow \boxed{\hat{v}_r = \frac{\hat{p}}{i\omega\rho_o} \left(ik - \frac{1}{r} \right)}$$

- ❑ We then obtain in the far-field ($kr \gg 1$):

$$\mathbf{I}_r = \frac{\Re(\hat{p}\hat{v}_r^*)}{2} \Rightarrow \mathbf{I}_r = \Re \left\{ \frac{\hat{p}\hat{p}^*}{2i\omega\rho_o} \left(ik + \frac{1}{r} \right) \right\} \Rightarrow \boxed{\mathbf{I}_r = \frac{k}{2\omega\rho_o} |\hat{p}|^2 = \frac{|\hat{p}|^2}{2c_o\rho_o}}$$

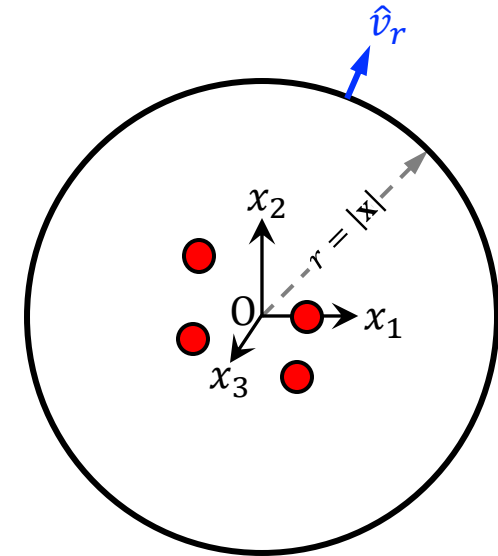
$$k = \frac{\omega}{c_o}$$

or

General signal

$$\boxed{\mathbf{I}_r = \frac{p_{\text{rms}}^2}{c_o\rho_o}}$$

$$p_{\text{rms}}^2 = |\hat{p}|^2 / \sqrt{2}$$



$$\boxed{\mathbf{I}_j = \frac{\Re(\hat{p}\hat{v}_j^*)}{2}}$$

Sound power of elementary sources

□ **Monopole** $\hat{p}(r) = -i\omega\rho_o\hat{Q}\frac{e^{ikr}}{4\pi r} \Rightarrow |\hat{p}|^2 = \hat{p}\hat{p}^* = \frac{(\omega\rho_o\hat{Q})^2}{(4\pi r)^2}$

➤ $\mathbf{I}_r = \text{const. @ } r = \text{const.}$

➤ Thus, we obtain:

$$W_a = \frac{|\hat{p}|^2}{2c_o\rho_o} \int_S dS \Rightarrow \boxed{W_a^{(\text{monopole})} = \frac{(\omega\rho_o\hat{Q})^2}{8\pi\rho_o c_o}}$$

□ **Dipole** $\hat{p}(r) \approx ikd \cos\theta \left(\frac{i\omega\rho_o\hat{Q}e^{ikr}}{4\pi r} \right) \left(1 - \frac{1}{ikr} \right)$

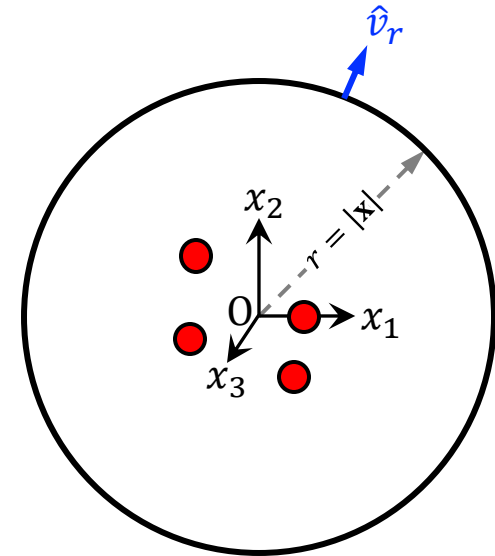
$$\Rightarrow |\hat{p}|^2 = (kd)^2 \cos^2\theta \left(\frac{\omega\rho_o\hat{Q}}{4\pi r} \right)^2 \left(1 + \frac{1}{(kr)^2} \right) \xrightarrow{\text{far-field, } kr \gg 1} |\hat{p}|^2 = (kd)^2 \cos^2\theta \left(\frac{\omega\rho_o\hat{Q}}{4\pi r} \right)^2$$

➤ $\mathbf{I}_r$ depends on $\cos^2\theta$

➤ Thus, we obtain: $W_a = \frac{|\hat{p}|^2}{2c_o\rho_o} \int_S dS \Rightarrow W_a = \frac{(kd)^2}{2c_o\rho_o} \left(\frac{\omega\rho_o\hat{Q}}{4\pi r} \right)^2 \int_0^\pi \int_0^{2\pi} r^2 \cos^2\theta \sin\theta d\theta d\varphi$

Two monopole sources in anti-phase emit noise an order of magnitude **less** than a single source

$$\Rightarrow \boxed{W_a^{(\text{dipole})} = \frac{(\omega\rho_o\hat{Q})^2 (kd)^2}{24\pi c_o\rho_o}} \quad \frac{(kd)^2}{3} \times W_a^{(\text{monopole})}$$



$$W_a = \int_S \mathbf{I} \cdot \mathbf{n}^{(o)} dS$$

$$\mathbf{I}_r = \frac{k}{2\omega\rho_o} |\hat{p}|^2 = \frac{|\hat{p}|^2}{2c_o\rho_o}$$

$$\int_S dS = \int_0^\pi \int_0^{2\pi} r^2 \sin\theta d\theta d\varphi$$

$$\int_0^\pi \cos^2\theta \sin\theta d\theta = \frac{2}{3}$$

Sound power of elementary sources

□ **Longitudinal quadrupole** $\hat{p}(r) \approx (kd)^2 \left(\frac{i\omega\rho_o\hat{Q}e^{ikr}}{4\pi r} \right) \cos^2 \theta$ Far-field, $kr \gg 1$

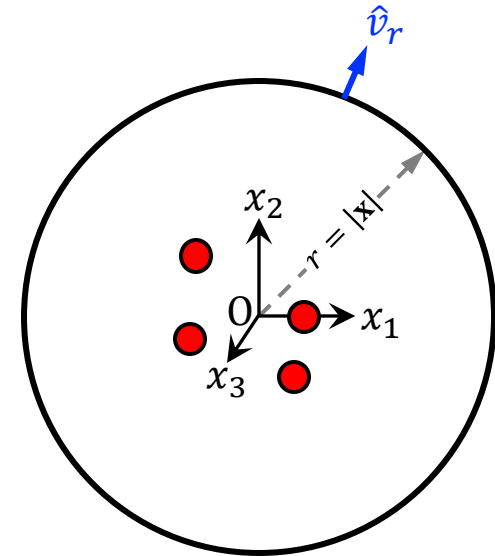
$$\Rightarrow |\hat{p}|^2 = (kd)^4 \cos^4 \theta \left(\frac{\omega\rho_o\hat{Q}}{4\pi r} \right)^2$$

➤ $\mathbf{I}_r$ depends on $\cos^4 \theta$

➤ Thus, we obtain:

$$W_a = \frac{|\hat{p}|^2}{2c_o\rho_o} \int_S dS \Rightarrow W_a = \frac{(kd)^4}{2c_o\rho_o} \left(\frac{\omega\rho_o\hat{Q}}{4\pi r} \right)^2 \int_0^\pi \int_0^{2\pi} r^2 \cos^4 \theta \sin \theta d\theta d\varphi$$

$$\Rightarrow W_a^{(long. quadrupole)} = \frac{(\omega\rho_o\hat{Q})^2 (kd)^4}{40\pi c_o\rho_o} \quad \frac{(kd)^4}{5} \times W_a^{(monopole)}$$



$$W_a = \int_S \mathbf{I} \cdot \mathbf{n}^{(o)} dS$$

$$\mathbf{I}_r = \frac{k}{2\omega\rho_o} |\hat{p}|^2 = \frac{|\hat{p}|^2}{2c_o\rho_o}$$

$$\int_S dS = \int_0^\pi \int_0^{2\pi} r^2 \sin \theta d\theta d\varphi$$

$$\int_0^\pi \cos^4 \theta \sin \theta d\theta = \frac{2}{5}$$

Sound power of elementary sources

□ Lateral quadrupole

$$\hat{p}(r) \approx (kd)^2 \left(\frac{i\omega\rho_o\hat{Q}e^{ikr}}{4\pi r} \right) \left(\frac{x_1x_2}{r^2} \right) \quad \text{Far-field, } kr \gg 1$$

$$\Rightarrow |\hat{p}|^2 = (kd)^4 \cos^2 \theta \sin^2 \theta \cos^2 \varphi \left(\frac{\omega\rho_o\hat{Q}}{4\pi r} \right)^2 \quad \frac{x_1x_2}{r^2} = \cos \theta \sin \theta \cos \varphi$$

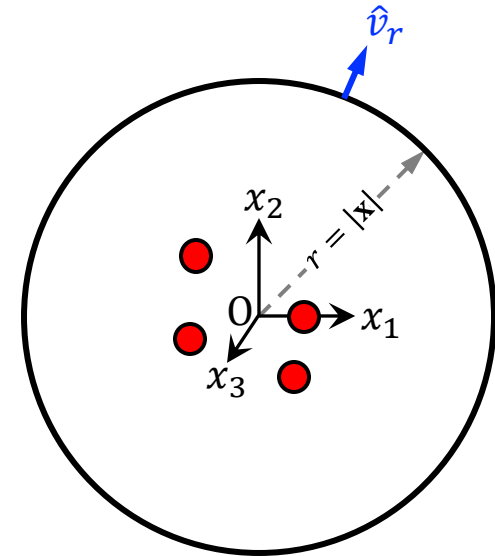
➤ $\mathbf{I}_r$ depends on $\cos^2 \theta \sin^2 \theta \cos^2 \varphi$

➤ Thus, we obtain:

$$W_a = \frac{|\hat{p}|^2}{2c_o\rho_o} \int_S dS \Rightarrow W_a = \frac{(kd)^4}{2c_o\rho_o} \left(\frac{\omega\rho_o\hat{Q}}{4\pi r} \right)^2 \int_0^\pi \int_0^{2\pi} r^2 \cos^2 \theta \sin^2 \theta \cos^2 \varphi \sin \theta d\theta d\varphi$$

$$\Rightarrow W_a^{(\text{lat. quadrupole})} = \frac{(\omega\rho_o\hat{Q})^2 (kd)^4}{120\pi c_o\rho_o} \quad \frac{(kd)^4}{15} \times W_a^{(\text{monopole})}$$

For a small compact acoustic source, the sound power output of the **dipole** and **quadrupole** sources is a factor of $(kd)^2/3$ and $(kd)^4/15$ less than the monopole source



$$W_a = \int_S \mathbf{I} \cdot \mathbf{n}^{(o)} dS$$

$$\mathbf{I}_r = \frac{k}{2\omega\rho_o} |\hat{p}|^2 = \frac{|\hat{p}|^2}{2c_o\rho_o}$$

$$\begin{aligned} \int_S dS &= \int_0^\pi \int_0^{2\pi} r^2 \sin \theta d\theta d\varphi \\ \int_0^\pi \cos^2 \theta \sin^3 \theta d\theta &= \frac{4}{15} \\ \int_0^{2\pi} \cos^2 \varphi d\varphi &= \pi \end{aligned}$$

Summary – compact sources acoustic efficiency

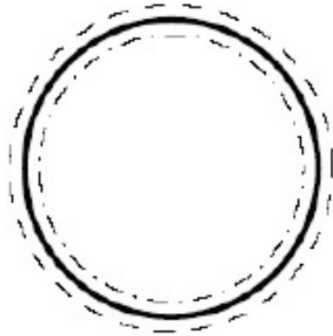
Efficient

Monopole

Dipole

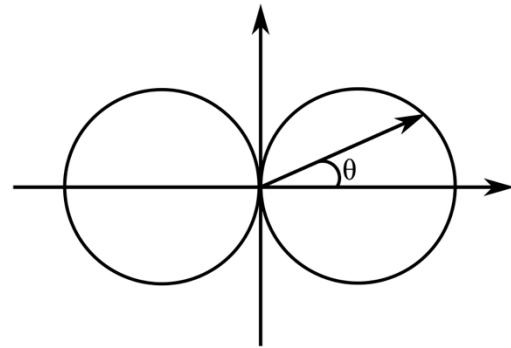
Longitudinal quadrupole

Lateral quadrupole



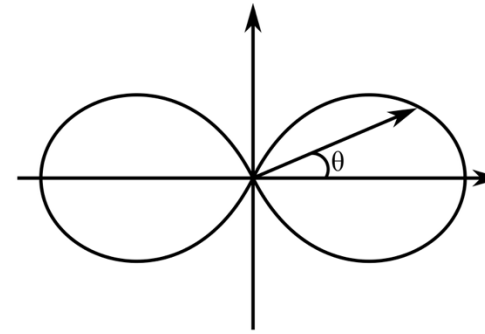
$$I \sim 1$$

$$W_a^{(mono.)} \sim 1$$



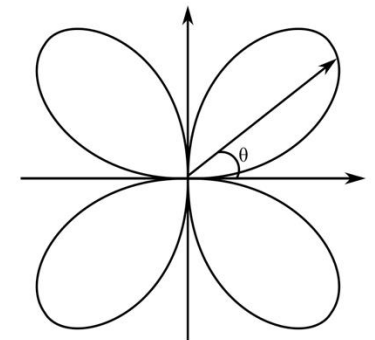
$$I \sim \cos^2 \theta$$

$$W_a^{(dipole)} \sim \frac{(kd)^2}{3} W_a^{(mono.)}$$



$$I \sim \cos^4 \theta$$

$$W_a^{(long. q.)} \sim \frac{(kd)^4}{5} W_a^{(mono.)}$$



$$I \sim \cos^2 \theta \sin^2 \theta \cos^2 \varphi$$

$$W_a^{(lat. q.)} \sim \frac{(kd)^4}{15} W_a^{(mono.)}$$

- For a small compact acoustic source ($kd \ll 1$), the sound power output of the **dipole** and **quadrupole** sources is a factor of $(kd)^2/3$ and $(kd)^4/5$ or $(kd)^4/15$ less than the monopole source

Sources with **no net volume displacement** are **inefficient acoustic sources**

In most aeroacoustics applications (flow-induced sound) – no mass is injected into the fluid...

Moving sources/observers

Assumptions

- Air at rest – $\mathbf{v}_o = 0$
- $\mathbf{y}$ – source position vector
- $\mathbf{v}_y$ – source velocity vector
- $\mathbf{x}$ – observer position vector
- $\mathbf{v}_x$ – observer velocity vector
- c_o – sound wave speed
- τ – sound source emission time – **retarded time**
- t – observer time when hearing the sound emitted at time τ
- $\hat{\mathbf{n}}_r$ – line-of-sight unit vector from *source* (at emission time τ) to *observer* (at reception t)

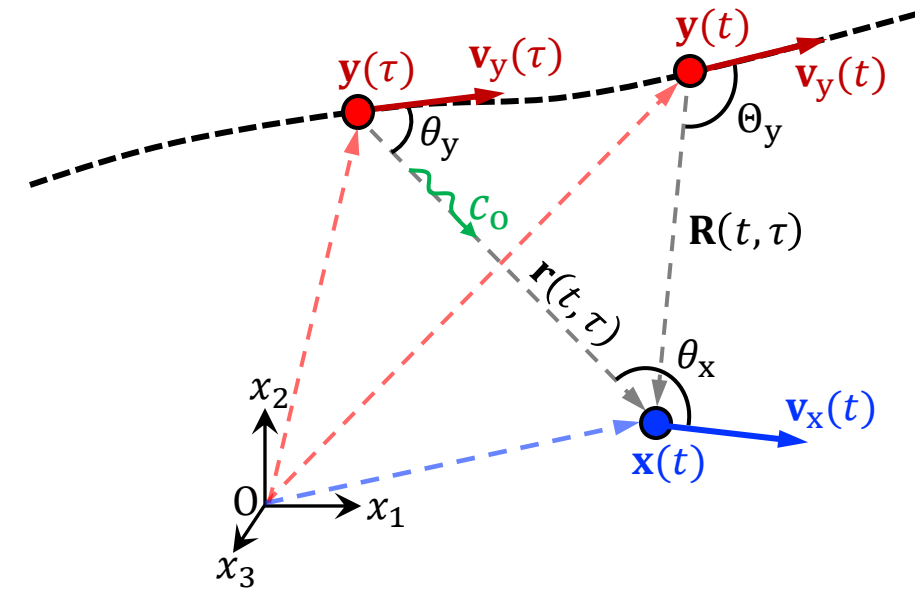
❑ A wavefront is emitted by the source at time τ , from position $\mathbf{y}(\tau)$, and is received by the observer at time t , at position $\mathbf{x}(t)$

❑ We can define the distance vector from the *source* to *observer* as: $\mathbf{r}(t, \tau) = \mathbf{x}(t) - \mathbf{y}(\tau)$

❑ Since sound propagates at speed c_o , we can also write that: $|\mathbf{r}(t, \tau)| = c_o \cdot (t - \tau) \equiv |\mathbf{x}(t) - \mathbf{y}(\tau)|$

❑ **Retarded time:**
$$\tau = t - \frac{|\mathbf{x}(t) - \mathbf{y}(\tau)|}{c_o}$$

Causality condition: $t > \tau$



Line-of-sight unit vector $\hat{\mathbf{n}}_r = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$

Moving sources/observers

- We shall now introduce the following function:

$$g(t, \tau) = t - \tau - \frac{|\mathbf{x}(t) - \mathbf{y}(\tau)|}{c_0} \equiv 0$$

- Based on causality ($t > \tau$), we consider that **t depends on τ** , $t = t(\tau)$

➤ A change in the emission time causes the reception time to change too

- We recognized that $g(t(\tau), \tau) = 0$, and can fully differentiate it g with respect to τ :

$$\frac{d}{d\tau} [g(t(\tau), \tau)] = \frac{\partial g}{\partial t} \frac{dt}{d\tau} + \frac{\partial g}{\partial \tau} = 0$$

- Computing the time derivative:

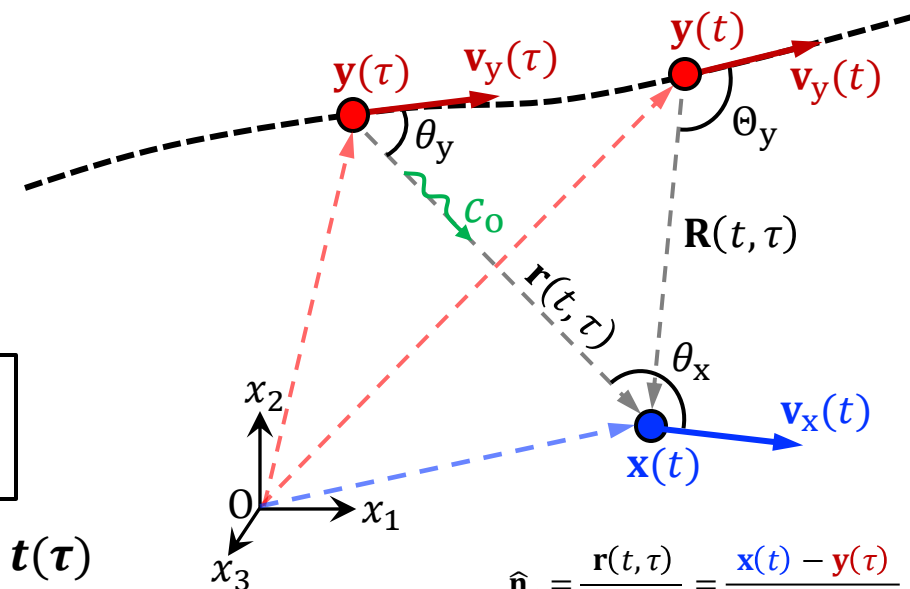
$$\frac{\partial g}{\partial t} = 1 - \frac{1}{c_0} \frac{\partial}{\partial t} |\mathbf{x}(t) - \mathbf{y}(\tau)| \Rightarrow \frac{\partial g}{\partial t} = 1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \underbrace{\frac{\partial}{\partial t} (\mathbf{x}(t) - \mathbf{y}(\tau))}_{= \mathbf{v}_x(t)} \Rightarrow$$

$$\frac{\partial g}{\partial t} = 1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_x(t) = 1 + M_x(t) \cos \theta_x$$

$\hat{\mathbf{n}}_r \cdot \mathbf{v}_x(t)$ – observer velocity in the source direction

$M_x(t) = \frac{|\mathbf{v}_x(t)|}{c_0}$ – observer Mach number

θ_x – angle between $\mathbf{v}_x$ and $\hat{\mathbf{n}}_r$



$$\hat{\mathbf{n}}_r = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

$$|\mathbf{r}(t, \tau)| = \hat{\mathbf{n}}_r \cdot \mathbf{r}(t, \tau)$$

Moving sources/observers

□ Similarly, the retarded time derivative is:

$$\frac{\partial g}{\partial \tau} = -1 - \frac{1}{c_0} \frac{\partial}{\partial \tau} |\mathbf{x}(t) - \mathbf{y}(\tau)| \Rightarrow \frac{\partial g}{\partial \tau} = 1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \underbrace{\frac{\partial}{\partial \tau} (\mathbf{x}(t) - \mathbf{y}(\tau))}_{= -\mathbf{v}_y(\tau)}$$

$$\Rightarrow \boxed{\frac{\partial g}{\partial \tau} = -1 + \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_y(\tau) = 1 - M_y(\tau) \cos \theta_y}$$

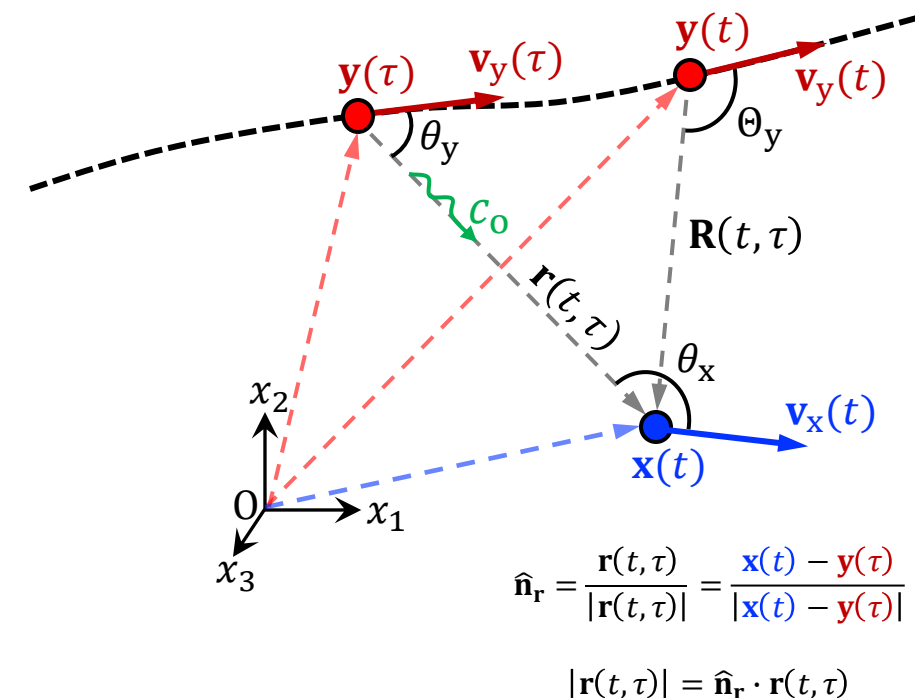
$\hat{\mathbf{n}}_r \cdot \mathbf{v}_y(\tau)$ – source velocity in the observer's direction

$M_y(\tau) = \frac{|\mathbf{v}_y(\tau)|}{c_0}$ – source Mach number

θ_y – angle between $\mathbf{v}_y$ and $\hat{\mathbf{n}}_r$

□ Substituting the above into $dg/d\tau$ yields:

$$\Rightarrow \boxed{\frac{dt}{d\tau} = \frac{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_y(\tau)}{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_x(t)} = \frac{1 - M_y(\tau) \cos \theta_y}{1 + M_x(t) \cos \theta_x}}$$



$$\frac{dg}{d\tau} = \frac{\partial g}{\partial t} \frac{dt}{d\tau} + \frac{\partial g}{\partial \tau} = 0 \Rightarrow \left(1 - \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_x(t) \right) \frac{dt}{d\tau} = - \left(-1 + \frac{\hat{\mathbf{n}}_r}{c_0} \cdot \mathbf{v}_y(\tau) \right)$$

Fundamental Kinematic Relation!

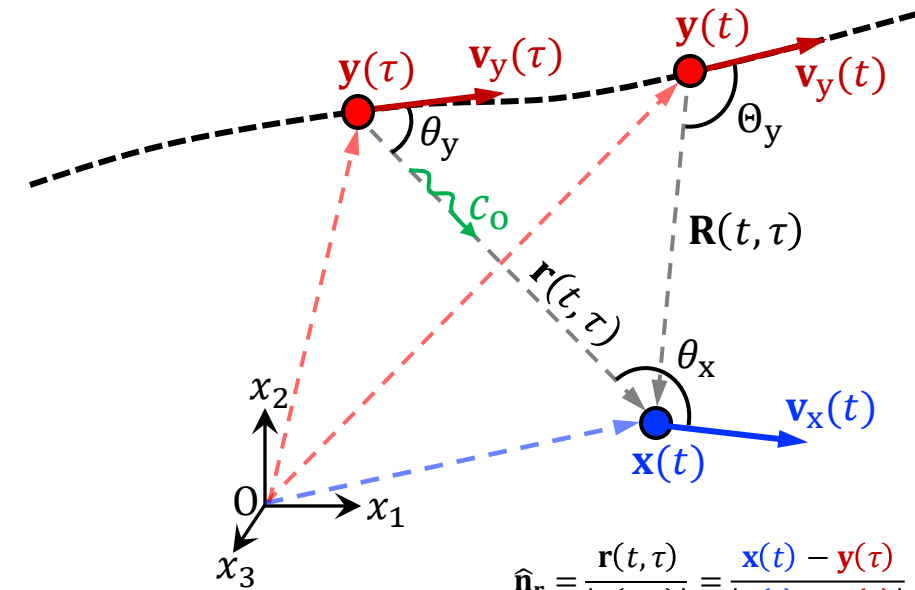
A small change in emission time $d\tau$ leads to a corresponding change in reception time dt

Moving sources/observers

- Assume a periodic sound signal is emitted with:
 - $T_y = d\tau$ – emission time period
 - $f_y = 1/d\tau$ – emitted frequency
- The observer received the sound signal with:
 - $T_x = dt$ – reception time period
 - $f_x = 1/dt$ – received frequency
- Therefore, we obtain the **Doppler frequency shift equation**:

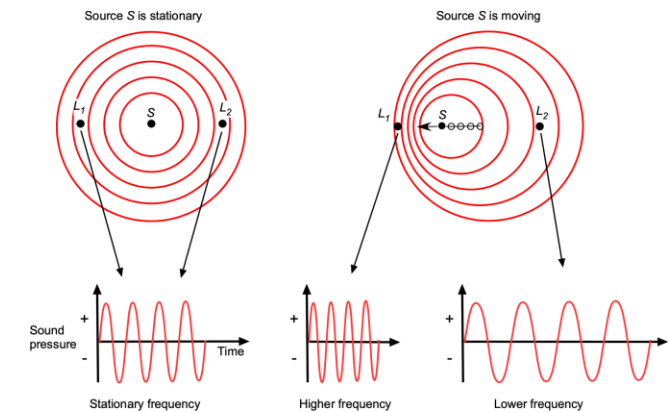
$$\frac{f_x}{f_y} = \frac{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_x(t)}{c_0 - \hat{\mathbf{n}}_r \cdot \mathbf{v}_y(\tau)} = \frac{c_0 + |\mathbf{v}_x(t)| \cos \theta_x}{c_0 - |\mathbf{v}_y(\tau)| \cos \theta_y} = \frac{1 + M_x(t) \cos \theta_x}{1 - M_y(\tau) \cos \theta_y}$$

- $|\mathbf{v}_x(t)| \cos \theta_x < 0 \rightarrow$ *observer* moves **away** from the *source*
- $|\mathbf{v}_y(\tau)| \cos \theta_y < 0 \rightarrow$ *source* moves **away** from the *observer*



$$\hat{\mathbf{n}}_r = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

$$|\mathbf{r}(t, \tau)| = \hat{\mathbf{n}}_r \cdot \mathbf{r}(t, \tau)$$



Moving sources/observers w. flow

- ❑ Consider now a **fluid medium with a uniform velocity $\mathbf{v}_0$**
- ❑ In this scenario, we shall consider two frames of reference:
 - **Stationary 'observer' frame $(\mathbf{x}, \mathbf{y}, t, \tau)$**
 - Fluid medium moves with constant **velocity $\mathbf{v}_0$**
 - **Moving fluid medium frame $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, \tilde{t}, \tilde{\tau})$**
 - Fluid medium is at rest ($\tilde{\mathbf{v}}_0 = 0$)
- ❑ The Galilean transformation between the two frames is:

$$\begin{aligned}\tilde{\mathbf{x}}(t) &= \mathbf{x}(t) - \mathbf{v}_0 t \\ \tilde{\mathbf{y}}(\tau) &= \mathbf{y}(\tau) - \mathbf{v}_0 \tau\end{aligned}$$

$$\begin{cases} t = \tilde{t} \\ \tau = \tilde{\tau} \end{cases}$$

Time is Galilean invariant...

$$\begin{cases} f_x = \tilde{f}_x \\ f_y = \tilde{f}_y \end{cases}$$

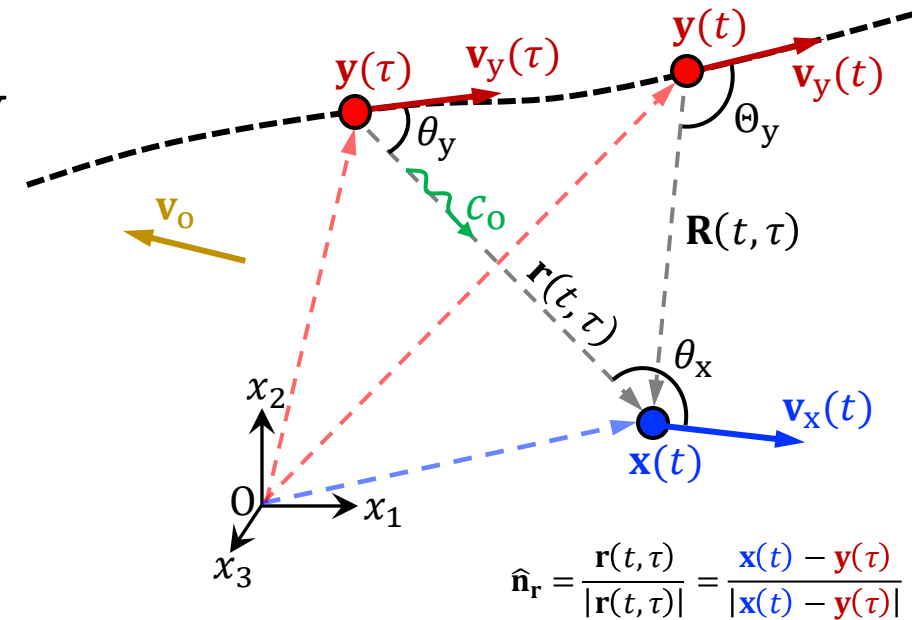
- ❑ In the *moving frame* ($\sim$) we can write:

$$\frac{d\tilde{\mathbf{x}}}{dt} = \frac{d}{dt}[\mathbf{x}(t) - \mathbf{v}_0 t] = \frac{d\mathbf{x}}{dt} - \mathbf{v}_0$$

$$\frac{d\tilde{\mathbf{y}}}{d\tau} = \frac{d}{d\tau}[\mathbf{y}(\tau) - \mathbf{v}_0 \tau] = \frac{d\mathbf{y}}{d\tau} - \mathbf{v}_0$$



$$\begin{aligned}\tilde{\mathbf{v}}_x(t) &= \mathbf{v}_x(t) - \mathbf{v}_0 \\ \tilde{\mathbf{v}}_y(\tau) &= \mathbf{v}_y(\tau) - \mathbf{v}_0\end{aligned}$$



$$|\mathbf{r}(t, \tau)| = \hat{\mathbf{n}}_r \cdot \mathbf{r}(t, \tau)$$

Moving sources/observers w. flow

- We repeat the same analysis we previously did for fluid at rest, but now we perform it in the 'new' *moving frame* to obtain:

$$\frac{dt}{d\tau} = \frac{c_0 - \widetilde{\mathbf{n}}_{\mathbf{r}} \cdot \widetilde{\mathbf{v}}_{\mathbf{y}}(\tau)}{c_0 - \widetilde{\mathbf{n}}_{\mathbf{r}} \cdot \widetilde{\mathbf{v}}_{\mathbf{x}}(t)}$$

- $\widetilde{\mathbf{n}}_{\mathbf{r}}$ – line-of-sight unit vector from **source** (τ) to **observer** (t) in the *moving frame*

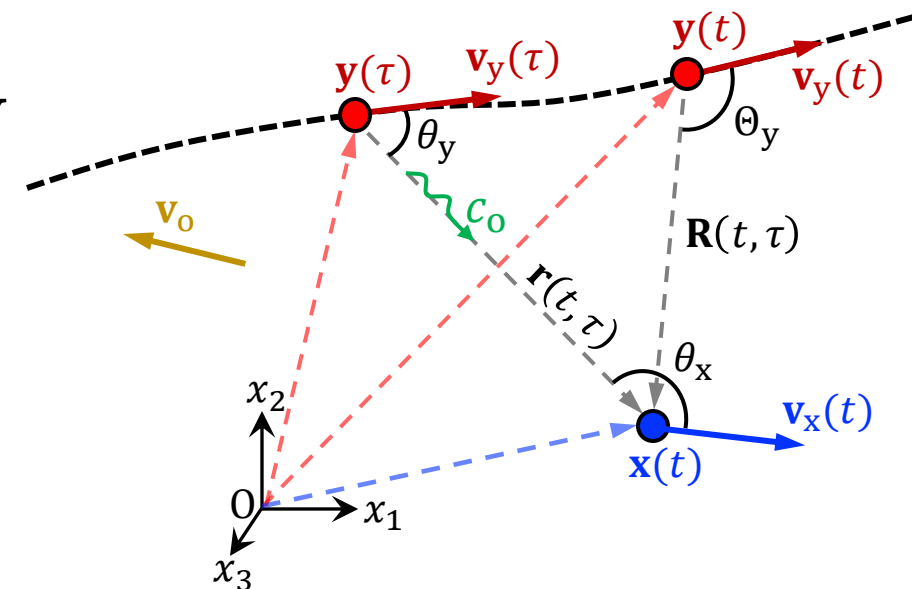
- Substituting back the stationary frame variables yields:

$$\begin{aligned} \widetilde{\mathbf{v}}_{\mathbf{x}}(t) &= \mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0 \\ \widetilde{\mathbf{v}}_{\mathbf{y}}(\tau) &= \mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0 \end{aligned} \Rightarrow$$

$$\frac{f_{\mathbf{x}}}{f_{\mathbf{y}}} = \frac{c_0 - \widetilde{\mathbf{n}}_{\mathbf{r}} \cdot (\mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0)}{c_0 - \widetilde{\mathbf{n}}_{\mathbf{r}} \cdot (\mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0)}$$

- For most **low Mach** ($M \ll 1$) uniform flows & **far-field observers** ($|\mathbf{r}(t, \tau)| \gg 1$) we can approximate:

$$\widetilde{\mathbf{n}}_{\mathbf{r}} \approx \widehat{\mathbf{n}}_{\mathbf{r}} \Rightarrow \frac{f_{\mathbf{x}}}{f_{\mathbf{y}}} \approx \frac{c_0 + |\mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0| \cos \tilde{\theta}_{\mathbf{x}}}{c_0 - |\mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0| \cos \tilde{\theta}_{\mathbf{y}}} = \frac{1 + M_{\mathbf{x}_r}(t) \cos \tilde{\theta}_{\mathbf{x}}}{1 - M_{\mathbf{y}_r}(\tau) \cos \tilde{\theta}_{\mathbf{y}}}$$



$$\widetilde{\mathbf{n}}_{\mathbf{r}} = \frac{\tilde{\mathbf{r}}(t, \tau)}{|\tilde{\mathbf{r}}(t, \tau)|} = \frac{\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)}{|\tilde{\mathbf{x}}(t) - \tilde{\mathbf{y}}(\tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_0(t - \tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau) - \mathbf{v}_0(t - \tau)|}$$

$$|\tilde{\mathbf{r}}(t, \tau)| = \widetilde{\mathbf{n}}_{\mathbf{r}} \cdot \tilde{\mathbf{r}}(t, \tau)$$

$$\widehat{\mathbf{n}}_{\mathbf{r}} = \frac{\mathbf{r}(t, \tau)}{|\mathbf{r}(t, \tau)|} = \frac{\mathbf{x}(t) - \mathbf{y}(\tau)}{|\mathbf{x}(t) - \mathbf{y}(\tau)|}$$

$$|\mathbf{r}(t, \tau)| = \widehat{\mathbf{n}}_{\mathbf{r}} \cdot \mathbf{r}(t, \tau)$$

$$M_{\mathbf{y}_r}(\tau) = \frac{|\mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0|}{c_0} \rightarrow \text{source Mach number relative to the flow}$$

$$M_{\mathbf{x}_r}(t) = \frac{|\mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0|}{c_0} \rightarrow \text{observer Mach number relative to the flow}$$

$$\tilde{\theta}_{\mathbf{y}} - \text{angle between } \mathbf{v}_{\mathbf{y}}(\tau) - \mathbf{v}_0 \text{ and } \widehat{\mathbf{n}}_{\mathbf{r}}$$

$$\tilde{\theta}_{\mathbf{x}} - \text{angle between } \mathbf{v}_{\mathbf{x}}(t) - \mathbf{v}_0 \text{ and } \widehat{\mathbf{n}}_{\mathbf{r}}$$